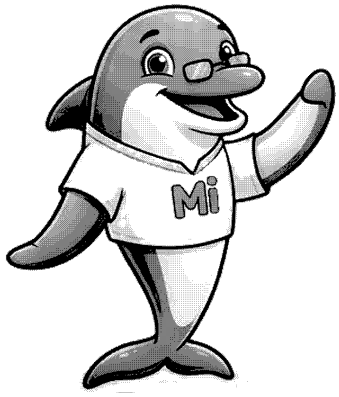
Math Inquirers

Extension Workbook

with answers

Level 5

Self-study | Ages 10-12



My name _____

About this book

Math Inquirers Extension Workbook, Level 5
Self-study practice with answers.

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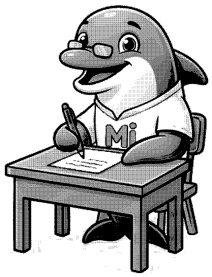
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- We have checked the answers with care. If you find a mistake, please tell us so we can fix it.
- Printed copies are black and white. The soft copy has the same pages in colour.

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Welcome!

This book is for you to explore at your own pace. Every unit has 7 pages. You can start on your own. A grown-up can help if you like.

How to use this book

1. **Read the Learn it page first. It shows how the idea works.**
2. **Do the Start here pages. Use a pencil. Mistakes help you learn.**
3. **Stuck? Find the hint. It is printed upside down.**
4. **Do Check yourself. Check the answers. Wrong? Read the Why.**

For the grown-up

- Practice for the same ideas as the Holiday Program (our school-holiday course), with new numbers and stories. You do not need to have joined it: every unit starts with an example.
- 3 short sessions a week, 20 to 25 minutes each, plus the weekend page **Go further** (see the pace plan).
- This is a self-study book. Your child marks their own work: each answer at the back says the answer first, then a short reason (Why) in plain words, and the page of the question. A child who reads slowly may like a helper to read the Why with them. The answers start on page 79: please ask your child to try first.
- Check yourself score 2 or less? Read the Learn it page again and redo one Start here page.
- Beyond Level 5 is ahead of school. You may skip it.

The four page names

- **Start here.** The main path. Do these first.
- **Think harder.** Longer problems. Hints are printed upside down.
- **Go further.** A weekend page. Big challenges, just for fun.
- **Check yourself.** Four quick questions at the end.

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My pace plan

Numbers in the table are page numbers. One unit a week for 10 weeks. Three short sessions a week, about 20 to 25 minutes each. **Go further** is a weekend page: do it on a weekend, when there is time to think. Tick the box when the week is done.

Week	Unit	Session 1 Learn it, Start here	Session 2 Start here	Session 3 Think harder, Check yourself	Weekend page Go further	Done
1	**Big number puzzles**	6-7	8-9	10 and 12	11	☐
2	**Fractions at work**	13-14	15-16	17 and 19	18	☐
3	**Change and compare**	20-21	22-23	24 and 26	25	☐
4	**Half the box**	27-28	29-30	31 and 33	32	☐
5	**Fill it up**	34-35	36-37	38 and 40	39	☐
6	**Rates and deals**	41-42	43-44	45 and 47	46	☐
7	**Out of 100**	48-49	50-51	52 and 54	53	☐
8	**Chase the angle**	55-56	57-58	59 and 61	60	☐
9	**Rules for patterns**	62-63	64-65	66 and 68	67	☐
10	**Pick a start**	69-70	71-72	73 and 75	74	☐
11	**Catch-up week**	Repeat any Check yourself where your score was 2 or less.				☐

Tips for the grown-up

- Short and often beats one long session. Stop while your child still wants more.
- Go further has big challenges. It is fine to skip it, or to do only one question. Missed a week? Carry on.

Learn it: Numbers to 10 million

My goal. I can read, compare and round numbers up to 10 million. I can test both sides of a target and check every clue.

Read a big number in groups of three from the right. To find the number closest to a target, test the best number just below it and just above it.

Worked example

Use the cards to make a 7-digit number as close as you can to 3 000 000. Use every card once. A number cannot start with 0.



1. **Just below the target: start with 2. Then put the cards in order from greatest to smallest: 2 985 310.**
2. **Just above the target: start with 3. Then put the rest in order from smallest to greatest: 3 012 589.**
3. **Find each gap. 3 000 000 - 2 985 310 = 14 690. And 3 012 589 - 3 000 000 = 12 589.**
4. **Compare the gaps. 12 589 is smaller than 14 690, so the number above is closer.**

Answer: The closest number is 3 012 589.

Closest number

1. Write the best number just below the target.
2. Write the best number just above the target.
3. Find the gap for each. The smaller gap wins.

Place-value chart

M	HTh	TTh	Th	H	T	O
4	0	5	3	2	0	1

- The number is 4 053 201.
- M is millions. TTh is ten-thousands.
- Rounding: a 5 in the next place rounds up.

Your turn

Answer at the back of the book.

Try **Use the cards to make a 7-digit number as close as you can to 8 000 000. Use every card once. Write your number.** _____



Start here

Do these first. Take your time.

1 **Fill in the blanks.**

(a) Write the number: 9 million, 40 thousand and 7.

_____

(b) In 3 470 218, the digit 7 stands for _____

(c) How many hundreds make 5 300 000? ☐ hundreds

2 **Arrange the numbers in order. Begin with the smallest.**
4 096 213, 940 621, 4 906 213, 4 096 231

_____

3 **Round 7 849 520 to the nearest:**

(a) thousand. _____

(b) hundred thousand. _____

(c) million. _____

4 **Look at the cards. Use all 7 cards once in each answer. A number cannot start with 0.**



(a) The greatest multiple of 5 is _____

(b) The smallest number is _____

5 **Mei has the number below. She adds 3 hundred-thousands. Then she takes away 6 ten-thousands. What is Mei's new number?**

M	HTh	TTh	Th	H	T	O
5	7	3	1	9	4	6

_____

Start here

Keep going. Check your work.

6 **Round each number to 1 digit to check. Circle one choice for each.**

(a) 3 985 × 52 = 20 722 could be right must be wrong

(b) 4 012 018 + 2 899 987 = 6 912 005 could be right must be wrong

(c) 8 120 × 295 = 2 395 400 could be right must be wrong

(d) 6 480 ÷ 16 = 4 050 could be right must be wrong

7 **Add ONE pair of brackets to each line to make it true. Write the new line.**

(a) 9 - 2 × 3 + 1 = 22 _____

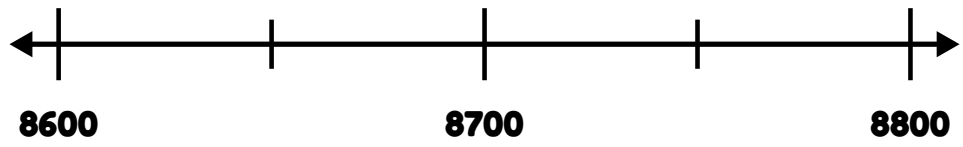
(b) 9 - 2 × 3 + 1 = 1 _____

(c) 2 400 000 ÷ 300 + 100 × 2 = 12 000

_____

8 **A factory makes 1 250 000 bolts. It packs 500 bolts in each box. Each carton holds 24 boxes. How many full cartons can it fill?** ☐ **cartons**

9 **A number rounded to the nearest hundred is 8 700. Fill in the blanks.**



(a) The smallest such number is ☐

(b) The greatest such number is ☐

10 **Circle True or False.**

(a) 5 400 000 is 54 hundred thousands. True False

(b) 1 000 000 - 1 = 99 999. True False

(c) The greatest 6-digit number is 1 less than the smallest 7-digit number.
True False

Start here

Nearly there. You can do it.

11 **Look at the table.**

Town	People
Alder	1 204 500
Birch	1 240 500
Cedar	1 024 500
Dune	1 420 050

(a) Which town has the most people? ☐

(b) How many more people live in the town with the most people than in the town with the fewest? _____

12 **Solve the riddle in the box. Check every clue when you finish.**

Clue 1 I am a 6-digit number. All my digits are different.

Clue 2 I round to 470 000 when rounded to the nearest ten-thousand.

Clue 3 My thousands digit is 2.

Clue 4 My hundreds digit is twice my tens digit.

Clue 5 My ones digit is 0.

I am _____

13 **A stadium has 48 000 seats. On Monday 3 100 people came. On Tuesday twice as many came. On Wednesday 1 250 fewer people came than on Tuesday. How many people came in all?** _____ **people**

14 **Which of these round to 5 000 000 to the nearest million? Circle them.**

4 499 999 4 500 000 5 499 999 5 500 000

Think harder

Hints are upside down. Try first!

1 **Solve the riddle in the box. Test the clues one at a time.**

Clue 1	**I am a 7-digit number. All my digits are different.**
Clue 2	**I round to 6 400 000 when rounded to the nearest hundred-thousand.**
Clue 3	**My ten-thousands digit is 1 less than my hundred-thousands digit.**
Clue 4	**My hundreds digit is 2 more than my thousands digit.**
Clue 5	**My tens digit is 0.**
Clue 6	**My digits add up to 33.**

I am _____

Hint. Clues 2 and 3 fix the first 3 digits. Then use Clue 6.

2 **Every number from 195 000 to 204 999 rounds to 200 000 when rounded to the nearest ten-thousand.**

(a) How many of these numbers have a digit sum of 4? ☐

Working: _____

Hint. Numbers from 195 000 to 199 999 have a digit sum over 4.

3 **Use each card once to make a 3-digit number and a 2-digit number. Multiply the 2 numbers. Make the product as great as you can.**



The product is _____

Working: _____

Hint. The greatest digits go first. Try both ways.

★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 5: Counting choices

1 **The first digit is never 0. How many 7-digit numbers have 7 different digits?**

_____

Working: _____

_____

_____

Hint. Count the choices for the first digit, then the second.

2 **Ali writes down every whole number from 1 to 999. He adds up all the digits he wrote. What is the total?** _____

Working: _____

_____

_____

Hint. Write 7 as 007. Each digit 0 to 9 appears equally often.

3 **Use each of 3, 4, 7 and 9 once. You may use + − × ÷ and brackets. Do not join digits. What is the number closest to 100 that you can make?** ☐

Working: _____

_____

_____

Hint. Try multiplying first. Which product is near 100?

4 **A printing press prints 2 400 sheets in 1 hour. It runs for 8 hours each day. A shop needs 1 000 000 sheets. How many days does the press need?** ☐ **days**

Working: _____

_____

_____

Hint. Sheets in one day first. Then divide. Is the answer a whole number?

✔ Check yourself

No hints. Try each one on your own.

1 **Look at the number 5 280 436.**

(a) The digit 2 stands for _____

(b) Rounded to the nearest 100 000, it is _____

Working: _____

2 **Build the 7-digit number nearest to 4 000 000. Use each card once.**



My number is _____

3 **Add ONE pair of brackets to 8 + 12 ÷ 4 × 2 = 10 to make it true. Write the new line.**

_____

Working: _____

4 **18 buses each carry 1 250 passengers in a week. How many passengers is that in 12 weeks?** _____

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

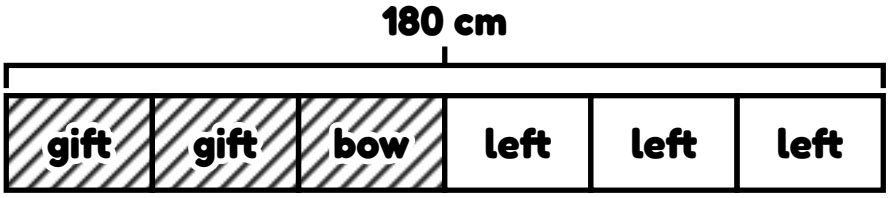
Learn it: Fractions: four operations

My goal. I can link fractions and division. I can find a fraction of a remainder, and work backwards with a bar model.

A fraction of a remainder is taken from what is left, not from the whole. First find what is left. Then take the fraction of that. A bar model with equal parts shows every step.

Worked example

A ribbon is 180 cm long. Mei uses 1/3 of it for a gift. Then she uses 1/4 of the ribbon that is left for a bow. How long is the ribbon that is left now?



1. Cut the bar into 6 equal parts. Each part is 180 ÷ 6 = 30 cm. The gift uses 1/3, which is 2 parts.
2. The part left is 4 parts. The bow uses 1/4 of those 4 parts, which is 1 part.
3. The ribbon that is left is 6 − 2 − 1 = 3 parts.
4. 3 parts × 30 cm = 90 cm.

Answer: The ribbon that is left is 90 cm long.

How to

1. Draw a bar. Cut it into equal small parts.
2. Take the first fraction. Find what is left.
3. Take the next fraction of what is left, not of the whole.

Remember

- a ÷ b = a/b. For example, 3 ÷ 4 = 3/4.
- A unit fraction has 1 on top, such as 1/5.
- To add or subtract, use the same bottom number.
- A fraction of a fraction: multiply the tops and the bottoms.

Your turn

Answer at the back of the book.

Try Mei has 120 beads. She uses 1/4 of them for a bracelet. Then she uses 2/3 of the rest for a necklace. How many beads are left? ☐

Start here

Do these first. Take your time.

1 **Fill in the blanks. Write each answer as a fraction.**

(a) 5 litres of juice fill 6 equal bottles. Each bottle holds ☐ litre.

(b) A 3 m rope is cut into 8 equal pieces. Each piece is ☐ m long.

(c) 9 ÷ 4 = ☐ (write a mixed number)

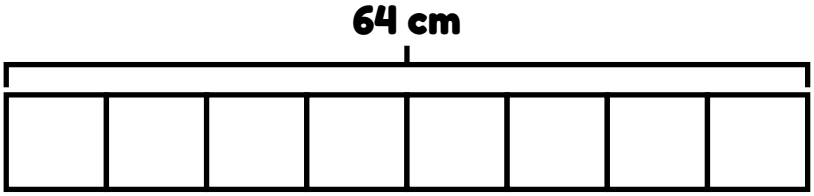
2 **Circle the greater amount in each pair.**

(a) 4/5 or 5/6 4/5 5/6

(b) 5/9 or 4/7 5/9 4/7

(c) 3/4 of 20 or 2/3 of 24 3/4 of 20 2/3 of 24

3 **A bar of 8 equal parts is 64 cm long. Find the fractions of the bar.**



(a) 3/8 of 64 cm is ☐ cm.

(b) 5/8 of 64 cm is ☐ cm.

(c) 1/4 of 64 cm is ☐ cm.

4 **Find each answer. Write it in its simplest form.**

(a) 2/3 + 1/4 = ☐

(b) 5/6 - 3/8 = ☐

(c) 3/4 - 1/6 = ☐

Start here

Keep going. Check your work.

5 Ravi walks 3/4 km each day. He does this for 6 days. How far does he walk in all? Write a mixed number. ☐ km

6 Sam has 72 cards. He gives 3/8 of them to Mei. Then he gives 2/3 of the cards that are left to Ben. How many cards does Sam have now? ☐

7 Write 7/12 as a sum of 2 different unit fractions. A unit fraction has 1 on top.

Take away 1/3 first: 7/12 − 1/3 = ☐ (simplest form). So 7/12 = 1/3 + ☐.

8 Look at the pattern: 1/3, 2/5, 3/7, 4/9, …

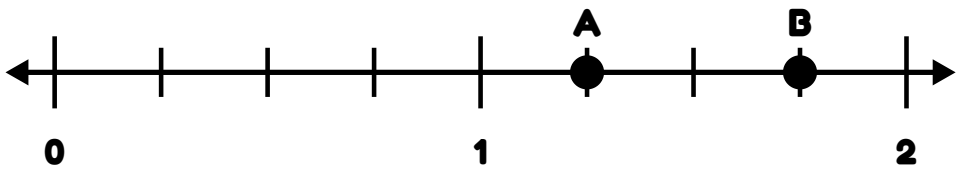
(a) The next 2 fractions are ☐ and ☐.

(b) The 10th fraction is ☐.

(c) 13/27 is the ☐ th fraction.

9 Ali reads 2/5 of a book on Saturday and 1/4 of the book on Sunday. What fraction of the book is not read yet? ☐

10 Read the points A and B on the number line. Write each as a mixed number.



(a) A = ☐

(b) B = ☐

Start here

Nearly there. You can do it.

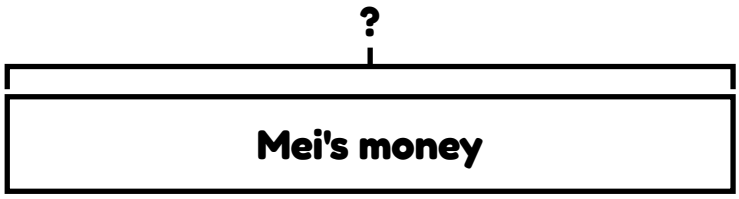
11 **Multiply. Write each answer in its simplest form.**

(a) 2/3 × 3/8 = ☐

(b) 3/5 of 2/3 = ☐

(c) 4/9 × 3 = ☐ (mixed number)

12 **Mei buys a book with 1/3 of her money. Then she pays 1/2 of the rest for lunch. She has $18 left. Cut the bar to show each step.**



(a) Lunch cost ☐ dollars.

(b) Mei had ☐ dollars at first.

Working: _____

13 **In a group of 40 children, 3/8 walk to school and 1/4 take the bus. The rest cycle.**

How many children cycle? ☐

Working: _____

14 **Circle True or False.**

(a) 3 ÷ 5 = 5/3 True False

(b) 7/8 is greater than 5/6. True False

(c) 1/4 of 1/2 is 1/8. True False

Think harder

Hints are upside down. Try first!

1 **Write 2/5 as the sum of 2 different unit fractions. Test each unit fraction that is less than 2/5 but greater than 1/5.**

2/5 = ☐ + ☐

Working: _____

_____

_____

Hint. Take away a unit fraction. Is what is left a unit fraction?

2 **A shop sells 1/3 of its eggs in the morning. It sells 1/4 of the rest in the afternoon. In the evening it sells 2/5 of the eggs then left. At the end 36 eggs are left. How many eggs did the shop have at first?** ☐

Working: _____

_____

_____

Hint. Find the fraction of ALL the eggs that is left at each step.

3 **Look at the cards. Make 2 fractions and add them. Each card is used once. The fractions need not be proper.**



(a) **The greatest sum is** ☐ **(mixed number)**

(b) **The smallest sum is** ☐

Working: _____

_____

_____

Hint. A big top and a small bottom make a big fraction.

★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 5: Dividing a fraction

1 **5/6 of a cake is shared equally among 10 children. What fraction of the whole cake does each child get?** ☐

Working: _____

_____

Hint. Draw the cake in 6 equal parts. Then think about sharing among 10.

2 **Write 1/6 as the sum of 2 different unit fractions. How many ways are there?**

☐ **ways**

Working: _____

_____

Hint. Start with 1/7. Is what is left a unit fraction?

3 **A ball falls from a height. Each time it bounces up to 3/4 of the height it fell from. After the 3rd bounce it reaches 27 cm.**

(a) The ball fell from ☐ cm at first.

(b) The 3rd bounce is ☐ cm lower than the 1st bounce.

Working: _____

Hint. Work backwards from 27 cm. What was the height before?

4 **Find 1/3 + 1/15 + 1/35 + 1/63.** ☐

Working: _____

_____

Hint. Can each fraction be written as a difference of 2 fractions?

✔ Check yourself

No hints. Try each one on your own.

1 **4 loaves are shared equally among 9 children. Write each answer as a fraction or mixed number.**

(a) Each child gets ☐ loaf. **(b)** 7 ÷ 3 = ☐

Working: _____

2 **Find each answer in its simplest form.**

(a) 7/8 − 1/3 = ☐ **(b)** 2/5 + 3/10 = ☐

Working: _____

3 **Kumar has 120 marbles. He gives 1/5 of them to Ali. He gives 3/8 of the rest to Wei. How many marbles does Kumar have now?** ☐

Working: _____

4 **Siti uses 3/4 of a bag of flour for bread and 1/6 of the bag for a cake. What fraction of the bag is left?** ☐

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

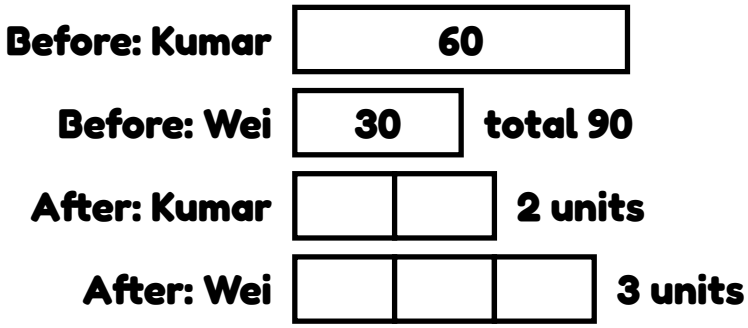
Learn it: Fraction word problems

My goal. I can use bar models for before and after. I ask what stays the same: the total, the difference or one amount.

Draw a model for the Before and a model for the After. Ask first: what stays the same?

Worked example

Kumar has 60 cards. Wei has 30 cards. Kumar gives some cards to Wei. Then Kumar has 2/3 as many cards as Wei. How many cards does Kumar give?



1. Kumar gives cards to Wei, so the total stays the same: 60 + 30 = 90.
2. After: Kumar has 2 units and Wei has 3 units. That is 5 units in all.
3. 5 units = 90, so 1 unit = 18. Kumar now has 2 × 18 = 36.
4. Kumar gives 60 − 36 = 24 cards.

Answer: Kumar gives 24 cards.

How to

1. Draw the Before bars. Draw the After bars.
2. Ask what stays the same. Make that part match.
3. Find 1 unit. Then answer the question.

What stays the same?

- A gives to B: the total.
- Both get or lose the same: the difference.
- Only one thing changes: the other amount.

Your turn

Answer at the back of the book.

Try **Jason has 70 stickers. Amelia has 50 stickers. Jason gives some stickers to Amelia. Then Jason has 3/5 as many stickers as Amelia. How many stickers does Jason give?** ☐

Start here

Do these first. Take your time.

1 **Nina has $96. A hat takes 3/8 of her money. A scarf takes 2/5 of the money then left.**

(a) The hat costs $ ☐

(b) The scarf costs $ ☐

(c) Nina has $ ☐ left.

2 **Lily has 22 counters. Ravi has 14 counters. Lily gives 5 counters to Ravi.**

(a) Lily has ☐ counters now.

(b) Ravi has ☐ counters now.

(c) The total is ☐ before and after.

3 **Dev is 9 years old. His uncle is 38 years old.**

(a) In 8 years Dev will be ☐ and his uncle will be ☐.

(b) Now the uncle is ☐ years older than Dev. In 8 years he is ☐ years older.

4 **One shelf has 52 books. Another shelf has 34 books. Some books are moved from the first shelf to the second. Then each shelf holds an equal number. How many books are moved?** ☐

5 **A jar holds 72 sweets. 5/12 of them are mints. How many more sweets are not mints than mints?** ☐

Start here

Keep going. Check your work.

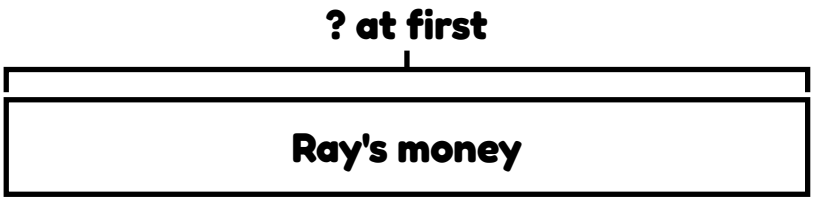
6 **A box has 12 red cubes and 8 blue cubes. Then 10 more red cubes are put in.**

(a) The cubes that did not change are red blue

(b) At first, blue cubes are ☐ of the cubes. (simplest form)

(c) Now, blue cubes are ☐ of the cubes. (simplest form)

7 **Ray spent 2/5 of his money on a bag. He has $45 left. Cut the bar to show each step.**



(a) 1 part is $ ☐

(b) Ray had $ ☐ at first.

8 **A pen costs 1/3 as much as a book. The pen and the book cost $28 together.**

(a) The book costs $ ☐

(b) The pen costs $ ☐

9 **Mei has 3/4 as many stickers as Siti. They have 56 stickers altogether.**

(a) Siti has ☐ stickers.

(b) Mei has ☐ stickers.

Start here

Nearly there. You can do it.

10 **What stays the same? Circle one choice for each.**

(a) Ali gives some marbles to Ben. the total the difference

(b) Ravi and Ali are each given $5. the total the difference

(c) Red beads are added to a jar of red and blue beads.
the red beads the blue beads

11 **Dev has $56. Mei has $29. Their uncle gives each child $12.**

(a) Dev has $ ☐ now.

(b) Mei has $ ☐ now.

(c) Dev has $ ☐ more than Mei now.

12 **Hana buys a shirt with 1/5 of her money. Shoes take 3/8 of the rest. She keeps $40. How much money did Hana start with? $** ☐

Working: _____

13 **A jug was 5/6 full of juice. After 250 ml was poured out, it was 1/2 full. How much juice does the jug hold when it is full?** ☐ **ml**

Working: _____

14 **Circle True or False.**

(a) When Ali gives Ben some stickers, the total number of stickers changes.
True False

(b) When 2 children are each given $5, the difference between their money stays the same. True False

(c) If Mei has 1/2 as many cards as Ali, then Ali has 2 times as many cards as Mei. True False

Think harder

Hints are upside down. Try first!

1 **On Monday Ravi spends 3/8 of his money. On Tuesday he spends 2/5 of what is left. After Tuesday he has $60. How much money did Ravi start with? $ ☐**

Working:

Hint. Find what fraction of all the money is left at the end.

2 **Ali and Ben have 132 cards altogether. Ben gives 14 cards to Ali. Then Ben has 4/7 of Ali's number of cards. How many cards did Ben have at first? ☐**

Working:

Hint. The total stays the same. Find each child's cards after.

3 **Ravi is 42 years old. His son is 12 years old. In how many years will the son be 1/3 of his father's age? ☐ years**

Working:

Hint. The difference in their ages never changes. How many units is it?

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 **Ali, Ben and Cara have 150 stickers altogether. Ali has twice as many stickers as Ben. Ben has 10 more than Cara.**

(a) Cara has ☐ stickers.

(b) Ben has ☐ stickers.

(c) Ali has ☐ stickers.

Working: _____

Hint. Call Cara's stickers 1 unit. Draw bars for Ben and for Ali.

2 **In a hall, 2/5 of the people are children. Then 15 adults leave and 15 children come in. Now 1/2 of the people are children. How many people are in the hall?**

☐

Working: _____

Hint. The total stays the same. How much more of the total is children?

3 **A jar holds red beads and white beads. Red beads make up 2/7 of the jar. Then 24 red beads are taken out. Now red beads make up 1/6 of the jar. How many beads were in the jar at first?** ☐

Working: _____

Hint. Which beads do not change? Start with them.

4 **Ali and Ben have 120 marbles altogether. Ali gives 1/3 of his marbles to Ben. Then Ben has 3 times as many marbles as Ali. How many marbles did Ali have at first?** ☐

Working: _____

Hint. After giving, Ali has 2/3 of his marbles. Ben has 3 times that.

✔ Check yourself

No hints. Try each one on your own.

1 **Jenny pays 1/5 of her money for a book. A bag takes 1/2 of the rest. She keeps $24. How much money did Jenny start with? $ ☐**

Working: _____

2 **Jamie has 46 cards. Jason has 30 cards. Jamie gives some cards to Jason. Then they have equal numbers. How many cards does Jamie give? ☐**

Working: _____

3 **Amelia has $50. Ray has $30. They are each given the same amount. Then Ray has 3/4 as much as Amelia. How much is each given? $ ☐**

Working: _____

4 **A pond has 24 ducks and 16 geese. Then 24 more geese come to the pond.**

(a) The birds that did not change are ducks geese

(b) The fraction of birds that are ducks is now ☐

Working: _____

My score: ☐ out of 4. Check the answers at the back.

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

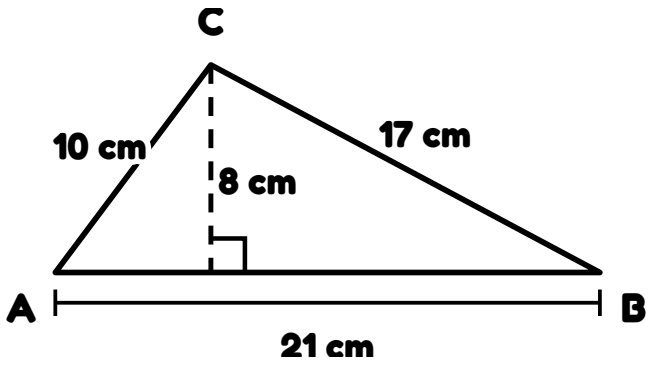
Learn it: Area of a triangle

My goal. I can find the area of a triangle with the height that meets the base at a right angle. I can find areas of shapes made from triangles.

The area of a triangle is 1/2 × base × height. The height meets the base at a right angle. A slanting side is not the height, unless the triangle has a right angle there.

Worked example

Find the area of triangle ABC. The dashed line is the height.



1. Choose the base: AB = 21 cm. The height meets AB at a right angle: it is 8 cm.
2. The sides AC = 10 cm and BC = 17 cm are slanting. They are not heights, so we do not use them.
3. Area = 1/2 × base × height = 1/2 × 21 × 8.
4. 21 × 8 = 168 and half of 168 is 84.

Answer: The area of triangle ABC is 84 cm².

How to

1. Find a base and the height that meets it at a right angle.
2. Ignore the slanting sides.
3. Multiply base × height. Then halve it.

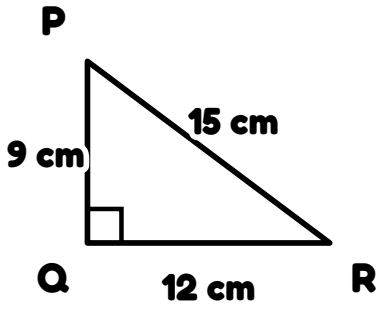
Your turn

Answer at the back of the book.

Try

Triangle PQR has a right angle at Q. PQ = 9 cm, QR = 12 cm and PR = 15 cm. Find its area.

_____ cm²



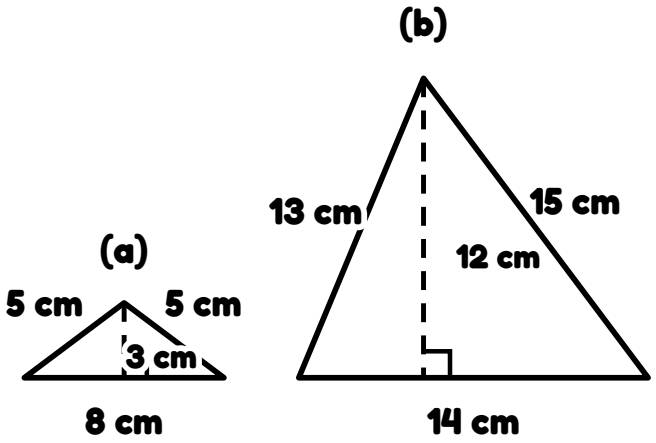
Start here

Do these first. Take your time.

1 **Find the area of each triangle. The dashed line is the height.**

(a) Triangle (a): ☐ cm²

(b) Triangle (b): ☐ cm²



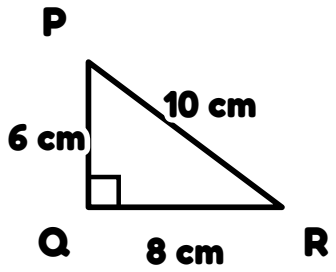
2 **Triangle PQR has a right angle at Q.**

(a) The area is ☐ cm².

(b) With QR as the base, the height is ☐ cm.

(c) With PQ as the base, the height is the side

QR PR



3 **The area of a triangle is 1/2 × base × height. Find the missing length.**

(a) Area 36 cm² and base 9 cm. The height is ☐ cm.

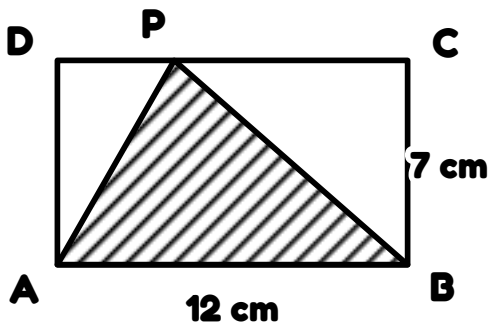
(b) Area 45 cm² and height 9 cm. The base is ☐ cm.

4 **ABCD is a rectangle. Point P lies on the top edge DC.**

(a) The shaded triangle ABP has area ☐ cm².

(b) P slides along DC. The area

gets greater | gets smaller | is the same

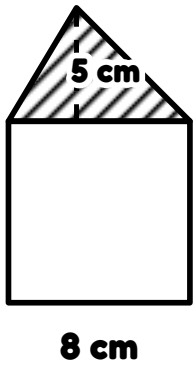


Start here

Keep going. Check your work.

5

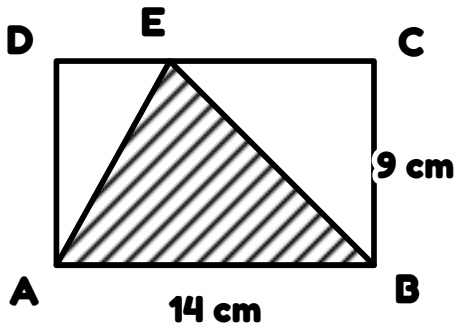
A square has sides of 8 cm. A triangle stands on its top edge, with the same 8 cm as its base. The triangle is 5 cm tall. What is the total area? ☐ **cm²**



6 **ABCD is a rectangle. Point E lies on the top edge DC. Triangle ABE is shaded.**

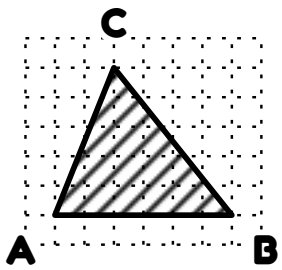
(a) The shaded area is ☐ cm².

(b) The area that is not shaded is ☐ cm².



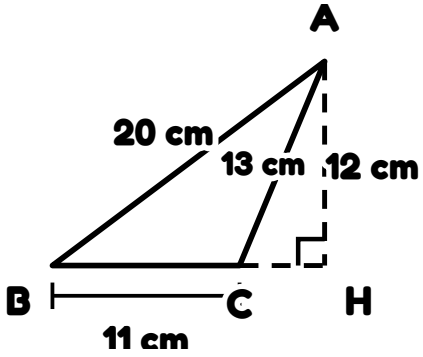
7

Each square on the grid is 1 cm². Find the area of the triangle. ☐ **cm²**



8

Look at triangle ABC. Its height AH is outside the triangle. Find the area of triangle ABC. ☐ **cm²**



Start here

Nearly there. You can do it.

9 **The areas of a square and a triangle are equal. The square has sides of 12 cm. The base of the triangle is 18 cm.**

(a) The area of the square is ☐ cm².

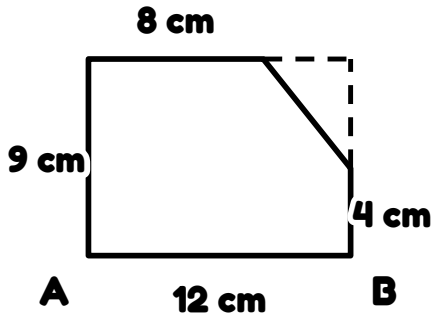
(b) The height of the triangle is ☐ cm.

Working: _____

10

The shape is a rectangle with a corner cut off.

Find its area. ☐ cm²



Working: _____

11 **A triangle has an area of 9 cm². Its base and height are whole numbers of cm.**

(a) How many pairs of base and height are possible? (Base 3 with height 6 is not the same pair as base 6 with height 3.) ☐

(b) The pair with a base of 3 cm has a height of ☐ cm.

Working: _____

12 **Circle True or False.**

(a) The height of a triangle meets its base at a right angle. True False

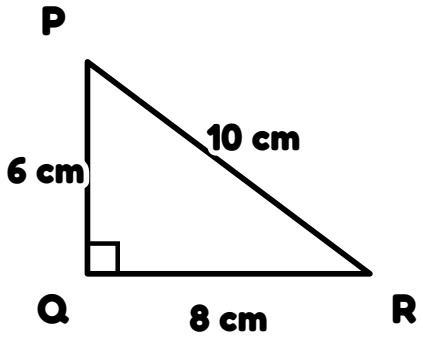
(b) Any slanting side of a triangle can be used as the height. True False

(c) A triangle with base 10 cm and height 6 cm has the same area as a triangle with base 6 cm and height 10 cm. True False

Think harder

Hints are upside down. Try first!

1 **Triangle PQR has a right angle at Q. Its sides are 6 cm, 8 cm and 10 cm. Now use the 10 cm side PR as the base. The height from Q to PR is ☐ cm.**

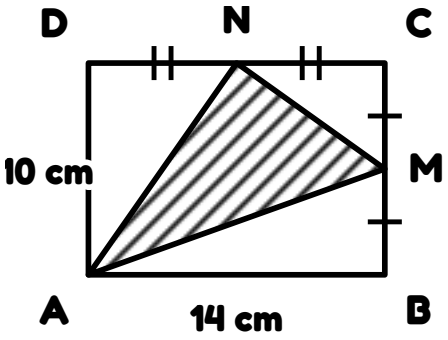


Working: _____

Hint. Find the area first. Then use the 10 cm side as the base.

2 **ABCD is a rectangle, 14 cm by 10 cm. M is the middle of BC. N is the middle of CD. How much area does the shaded triangle AMN cover?**

☐ **cm²**



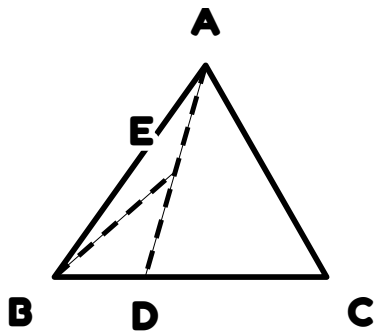
Working: _____

Hint. The shaded triangle is part of the rectangle. What is the rest?

3 **The area of triangle ABC is 72 cm². D is on BC and BD is 1/3 of BC. E is the middle of AD.**

(a) The area of triangle ABD is ☐ cm².

(b) The area of triangle BDE is ☐ cm².



Working: _____

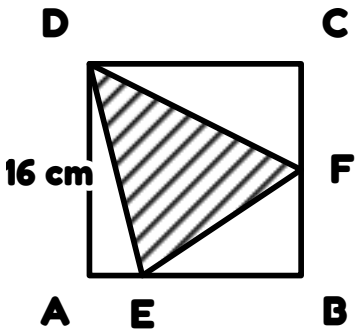
Hint. ABD and ABC have the same height. Compare their bases.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1

ABCD is a square with sides of 16 cm. E lies on AB, 4 cm from A. F is the middle of BC. What is the area of triangle DEF? ☐ **cm²**



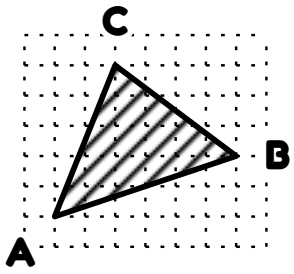
Working: _____

Hint. Which shapes are left when you remove triangle DEF?

2

Each square on the grid is 1 cm². The corners of the triangle are on grid points. Find its area.

☐ **cm²**



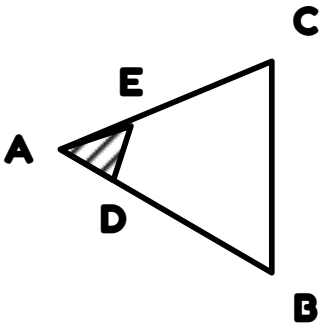
Working: _____

Hint. Draw a rectangle round the triangle. Take away 3 triangles.

★ **Beyond Level 5: Corner triangles**

3

Triangle ABC has an area of 96 cm². D is on AB with AD = 1/4 of AB. E is on AC with AE = 1/3 of AC. Find the area of triangle ADE. ☐ **cm²**



Working: _____

Hint. Draw one more line. Compare a larger triangle with ABC first.

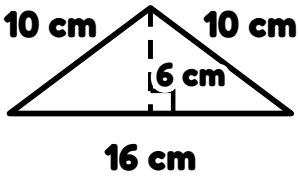
Name _____

✔ Check yourself

No hints. Try each one on your own.

1

Find the area of the triangle. ☐ **cm²**

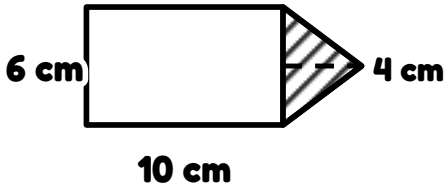


2 **A triangle has an area of 70 cm². Its base is 14 cm. Find its height.** ☐ **cm**

Working: _____

3

Find the area of the whole shape. ☐ **cm²**



4 **A garden is a triangle with a base of 18 m and a height of 10 m. 1 bag of seed covers 6 m². How many bags are needed to cover the whole garden?** ☐

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

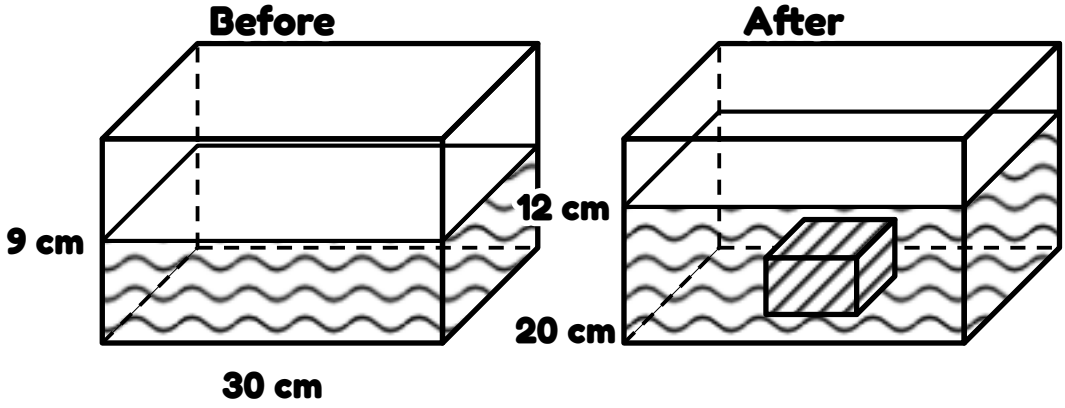
Learn it: Volume

My goal. I can find the volume of a cuboid and the volume of water in a tank. I can find the volume of a stone from how much the water rises.

Volume is the space inside a solid. For a cuboid, volume = length × breadth × height. 1 litre is 1000 cm³. A stone that sinks pushes the water up.

Worked example

The floor of a tank is 30 cm × 20 cm. Water fills it to a depth of 9 cm. A stone sinks and the level rises to 12 cm. Find the volume of the stone.



1. The water rises 12 − 9 = 3 cm. The stone pushes the water up.
2. The stone's volume equals the extra water, a block 30 × 20 × 3 cm.
3. Volume = 30 × 20 × 3. First 30 × 20 = 600.
4. 600 × 3 = 1800.

Answer: The volume of the stone is 1800 cm³.

How to

1. Find the rise: new depth − old depth.
2. Volume = base length × base width × rise.
3. Write the unit: cm³. 1000 cm³ = 1 litre.

Also remember

- Poured water keeps its volume.
- Depth = volume ÷ floor area.
- Packing: fit whole cubes along each edge.

Your turn

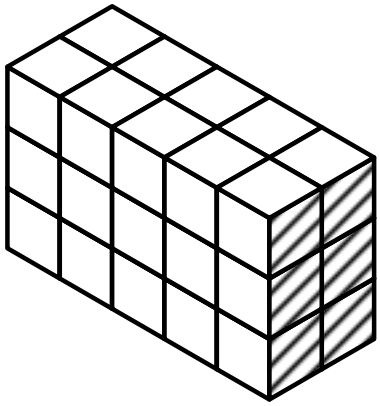
Answer at the back of the book.

Try **A tank floor is 25 cm × 20 cm. Water stands 10 cm deep. A stone of volume 1500 cm³ sinks in it. Find the new depth of the water.** ☐ cm

Start here

Do these first. Take your time.

1 **A box is built from 1 cm cubes. It is 5 cubes long, 2 cubes wide and 3 cubes high.**



(a) The front bottom row has ☐ cubes.

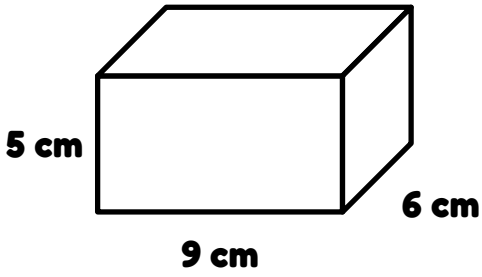
(b) The bottom layer has ☐ cubes.

(c) The box has ☐ cubes. Its volume is ☐ cm^3.

2 **A cuboid measures 9 cm × 6 cm × 5 cm.**

(a) The volume of the cuboid is ☐ cm^3.

(b) A cube has edges of 7 cm. Its volume is ☐ cm^3.



3 **Find the missing length of each cuboid.**

(a) The base is 8 cm by 5 cm. The volume is 200 cm^3. The height is ☐ cm.

(b) The base is a square. The height is 4 cm. The volume is 144 cm^3. Each side of the base is ☐ cm.

4 **Fill in the blanks. Remember 1 litre = 1000 ml = 1000 cm^3.**

(a) 3.5 litres = ☐ cm^3

(b) 12 000 cm^3 = ☐ litres

(c) A tank is 50 cm long, 40 cm wide and 30 cm high. It holds ☐ litres when full.

Start here

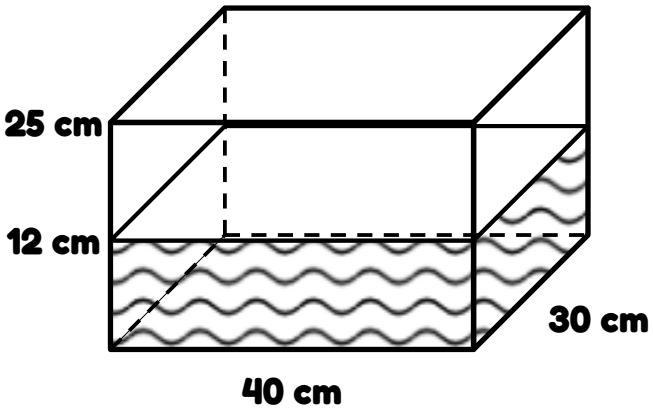
Keep going. Check your work.

5 **A tank measures 40 cm × 30 cm × 25 cm. It holds water 12 cm deep.**

(a) The water has a volume of ☐ cm^3.

(b) That is ☐ litres.

(c) To fill the tank to the top, ☐ litres more are needed.



6 **A tank floor is 40 cm × 25 cm. The water level is 7 cm. Then 5 litres are poured in.**

(a) The water rises ☐ cm.

(b) The water is now ☐ cm deep.

7 **The floor of a tank is 24 cm × 15 cm and the water is 8 cm deep. A stone sinks and the water level reaches 11 cm. Find the stone's volume.** ☐ cm^3

8 **A tank has a base 60 cm by 30 cm. The water is 20 cm deep. Then 9 litres of water are taken out.**

(a) The water level falls ☐ cm.

(b) The water is now ☐ cm deep.

9 **Cuboids are made with 30 cubes, using all the cubes each time. The edges are whole numbers of cm. A cuboid that is turned round is not new.**

(a) There are ☐ different cuboids.

(b) The cuboid with the smallest total of the 3 edges has a total of ☐ cm.

Start here

Nearly there. You can do it.

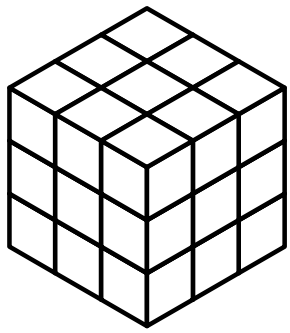
10 **A big cube is made from 27 small cubes (3 by 3 by 3). All its outside is painted. Then it is taken apart.**

(a) Cubes with 3 painted faces: ☐

(b) Cubes with 2 painted faces: ☐

(c) Cubes with 1 painted face: ☐

(d) Cubes with no paint: ☐



11 **Cubes with edges of 6 cm are packed in neat layers in a carton 32 cm long, 20 cm wide and 14 cm high. The cubes cannot be cut. How many cubes fit in the carton?** ☐

12 **A tub 45 cm long, 40 cm wide and 30 cm high is full of water. All of the water goes into an empty tank. The tank floor is 60 cm × 30 cm. Find the depth of the water in the tank.** ☐ **cm**

13 **A tank is 3/8 full of water. Then 35 litres more go in and the tank is 7/12 full. Find the capacity of the full tank.** ☐ **litres**

Working: _____

14 **Circle True or False.**

(a) 1 litre of water has a volume of 1000 cm^3. True False

(b) A cuboid 2 cm × 3 cm × 4 cm has the same volume as a cube with edges of 3 cm. True False

(c) If all 3 edges of a cuboid are doubled, its volume is doubled.
True False

Think harder

Hints are upside down. Try first!

1 **A big cube is made of small cubes. Its outside is painted. Then it is taken apart. 64 of the small cubes have no paint. How many small cubes make the big cube?**

☐

Working: _____

_____

_____

Hint. Think about the cubes with no paint. What shape do they make?

2 **Tank A has a floor 50 cm × 20 cm. Its water is 24 cm deep. Tank B has a floor 25 cm × 20 cm and starts empty. Water is poured from A into B until the 2 tanks have equal depths. Find that depth.** ☐ **cm**

Working: _____

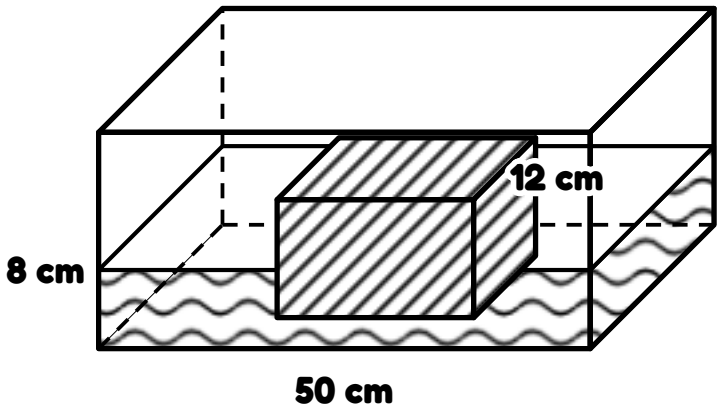
_____

_____

Hint. The same water now covers both floors. Add the floor areas.

3 **A tank floor is 50 cm × 30 cm and the water is 8 cm deep. A block 20 cm × 15 cm × 12 cm stands upright on the floor, with its 20 cm × 15 cm face down. Find the new water depth.**

☐ **cm**



Working: _____

Hint. Water cannot go where the block is. Find the free floor area.

★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 5: Cubes with tunnels

1 **A solid cube has edges of 6 cm. A tunnel 2 cm by 2 cm goes straight through the middle from the front to the back. 2 more tunnels of the same size go through the middle from left to right and from top to bottom. What is the volume of the solid that is left?** ☐ **cm³**

Working: _____

_____

_____

_____

Hint. Count the cubes in one tunnel. Then think about overlap.

2 **A tank measures 60 cm × 50 cm × 40 cm and is empty. A tap pours in 9 litres each minute. A leak loses 1 litre each minute. After how many minutes is the tank full?**

☐

Working: _____

_____

_____

_____

Hint. Find the capacity. Then the net litres gained each minute.

3 **The areas of 3 faces of a cuboid that meet at one corner are 20 cm², 28 cm² and 35 cm². Find the volume of the cuboid.** ☐ **cm³**

Working: _____

_____

_____

_____

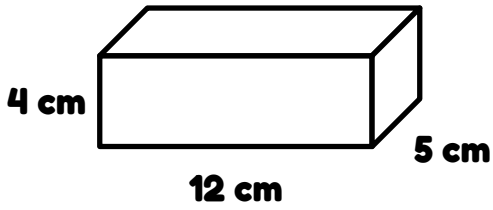
Hint. Try small whole-number edges. Which pairs multiply to 20?

✔ Check yourself

No hints. Try each one on your own.

1 **Find the volume of the cuboid.**

☐ cm³



2 **A tank floor is 50 cm × 30 cm and the tank holds 18 litres. Find the water depth.**

☐ cm

Working: _____

3 **A cube is made of 125 small cubes (5 by 5 by 5). Its outside is painted. Count the small cubes with exactly 2 painted faces.** ☐

Working: _____

4 **A tank floor is 20 cm × 15 cm and the water is 10 cm deep. A stone sinks and the water rises to 14 cm. Find the volume of the stone.** ☐ cm³

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

Learn it: Rate

My goal. I can find a rate for 1 unit and use it to scale up. I can compare buys, and add rates when taps or machines work together.

A rate tells how much for 1 unit, such as litres in 1 minute. Find the amount for 1 unit first. Then multiply to scale up, or divide to find how long.

Worked example

A printer prints 90 pages in 6 minutes. How many pages does it print in 15 minutes? How many minutes does it need for 300 pages?

Minutes	Pages
6	90
1	15
15	225
20	300

1. **Find the rate: 90 ÷ 6 = 15 pages in 1 minute.**
2. **In 15 minutes: 15 × 15 = 225 pages.**
3. **For 300 pages: 300 ÷ 15 = 20 minutes.**
4. **Check: 20 minutes × 15 pages = 300 pages.**

Answer: It prints 225 pages in 15 minutes. It needs 20 minutes for 300 pages.

How to

1. Find the amount for 1 unit.
2. Multiply to scale up. Divide to find how long.
3. Check with the other way round.

Buys and teams

- Better buy: find the cost of 1 item.
- Taps working together: add the rates.
- A leak: take its rate away.

Your turn

Answer at the back of the book.

Try **A machine fills 24 bottles in 4 minutes. How many bottles does it fill in 11 minutes?** ☐

Start here

Do these first. Take your time.

1 A tap fills 18 litres in 3 minutes. Fill in the table. Write the missing numbers.

Minutes	Litres
3	18
1	
5	
8	

2 6 pens cost $4.20.

(a) 1 pen costs $ ☐

(b) 9 pens cost $ ☐

3 An oven bakes 36 rolls in 4 minutes.

(a) It bakes ☐ rolls in 1 minute.

(b) In 25 minutes it bakes ☐ rolls.

(c) It needs ☐ minutes to bake 270 rolls.

4 Pack A has 5 pencils for $2.50. Pack B has 8 pencils for $3.60.

(a) Pack A costs $ ☐ for 1 pencil.

(b) Pack B costs $ ☐ for 1 pencil.

(c) The better buy is Pack A Pack B

Start here

Keep going. Check your work.

5 **Shop X sells 3 cans for $2.40. Shop Y sells 5 cans for $3.75. Mei needs 15 cans.**

(a) 15 cans at Shop X cost $ ☐

(b) 15 cans at Shop Y cost $ ☐

(c) The cheaper shop is Shop X Shop Y

6 **Tap A fills 14 litres a minute. Tap B fills 6 litres a minute. Both taps are on.**

(a) Together they fill ☐ litres in 1 minute.

(b) They need ☐ minutes to fill a 200 litre tank.

7 **A pump puts 25 litres a minute into a tank. A drain lets out 10 litres a minute. Both are on. In how many minutes is a 300 litre tank full?** ☐

Working: _____

8 **Machine P folds 35 booklets each minute. Machine Q folds 25 booklets each minute. Together they fold one order of 1200 booklets. How many minutes does the order take?** ☐

Working: _____

9 **A car park charges $3.00 for the first 2 hours. After that it charges $1.20 for each 15 minutes, and a part of 15 minutes counts as a whole one. Ravi parks for 3 hours 40 minutes. How much does he pay? $** ☐

Working: _____

Start here

Nearly there. You can do it.

10 **A tank has 240 litres of water. A pump takes out 15 litres each minute.**

(a) After 9 minutes, ☐ litres are left.

(b) The tank is empty after ☐ minutes.

11 **A tap drips 4 ml of water every 10 seconds.**

(a) In 1 minute it drips ☐ ml.

(b) In 1 hour it drips ☐ ml.

(c) That is ☐ litres in 1 hour.

12 **Circle True or False.**

(a) If 6 cans cost $3.60, then 12 cans cost $7.20. True False

(b) A larger pack is always the better buy. True False

(c) Tap A fills 5 litres a minute. Tap B fills 7 litres a minute. Together they fill 12 litres a minute. True False

13 **Gym plan A costs $18 a month and then $2 for each visit. Plan B costs $5 for each visit and has no monthly fee. For how many visits in a month do the 2 plans cost the same?** ☐

Working: _____

14 **A pipe delivers 1.5 litres of water every 20 seconds. How many litres does it deliver in 5 minutes?** ☐ **litres**

Working: _____

Think harder

Hints are upside down. Try first!

1 **Pen packs: A has 5 pens for $3.00, B has 8 pens for $4.70 and C has 12 pens for $6.90. Mei needs exactly 32 pens and may mix packs. Find the least total cost. $** ☐

Working: _____

_____

_____

_____

Hint. List every way to make 32 pens with whole packs.

2 **Alone, hose P needs 20 minutes to fill a pool. Alone, hose Q needs 30 minutes to fill the same pool. Find the time to fill the pool when both hoses are on.**

☐ **minutes**

Working: _____

_____

_____

_____

Hint. Pretend the pool holds a number that 20 and 30 both go into.

3 **5 workers pack 400 boxes in 8 minutes. All the workers pack at the same rate.**

How many boxes do 8 workers pack in 6 minutes? ☐

Working: _____

_____

_____

_____

Hint. Find the boxes for 1 worker in 1 minute first.

★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 5: Average rate

1 **A tank holds 120 litres. The first 60 litres go in at 6 litres a minute and the second 60 litres at 12 litres a minute. Average rate = total litres ÷ total minutes.**

Find the average rate for the whole tank. ☐ litres a minute

Working: _____

_____

_____

_____

Hint. Find the minutes for each half first. Then add the minutes.

2 **Tank A has 90 litres. A pump takes 5 litres out of it each minute. Tank B has 10 litres. A tap puts 3 litres into it each minute. After how many minutes do the 2 tanks have the same amount of water? ☐ minutes**

Working: _____

_____

_____

_____

Hint. How many litres does the gap between the tanks close each minute?

3 **Tap A fills 9 litres a minute. Tap B fills 11 litres a minute. Tap A is turned on at the start. Tap B is turned on 8 minutes later. How many minutes after the start is a 312 litre tank full? ☐ minutes**

Working: _____

_____

_____

_____

Hint. What has happened before tap B is turned on?

✔ Check yourself

No hints. Try each one on your own.

1 **A copier makes 54 copies in 3 minutes. How many in 8 minutes?** ☐

Working: _____

2 **5 mangoes cost $4.00. 8 mangoes cost $6.00.**

(a) **The better buy is** **5 mangoes** **8 mangoes**

(b) **It costs $** ☐ **for 1 mango.**

Working: _____

3 **Tap A fills 9 litres a minute, tap B fills 11 and a leak loses 4. All 3 work together. Find the time to fill 160 litres.** ☐ **min**

Working: _____

4 **Bike hire costs $5.00 for the first 30 minutes. After that it costs $2.00 for each 20 minutes, and a part of 20 minutes counts as a whole one. How much for 1 hour 55 minutes? $** ☐

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

Learn it: Percentage

My goal. I can find a percentage of an amount from 10%, 5% and 1%. I can find a discount and GST, and avoid percentage traps.

Percent means out of 100. Build other percentages from 10%, 5% and 1%. A discount comes off the first cost. GST is charged on the cost after the discount.

Worked example 1

A bag costs $60. It has 30% off. GST of 9% is charged on the cost after the discount. How much does Ali pay?



1. 10% of $60 = $6. So 30% = 3 × $6 = $18 off.
2. The cost after the discount is $60 − $18 = $42.
3. 1% of $42 = $0.42. So 9% = 9 × $0.42 = $3.78 GST.
4. Ali pays $42 + $3.78 = $45.78.

Answer: Ali pays $45.78.

Worked example 2

Find 35% of $160.

1. 10% of $160 = $16.
2. 5% is half of 10%, so 5% = $8.
3. 35% = 10% + 10% + 10% + 5% = $16 + $16 + $16 + $8 = $56.

Answer: 35% of $160 is $56.

How to

1. Find 10%. Build the percentage you need.
2. Take off the discount first.
3. Find the GST on the new amount.

Your turn

Answer at the back of the book.

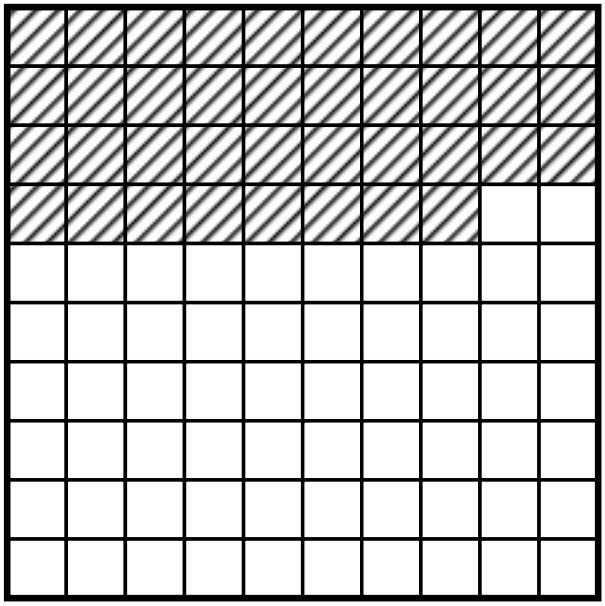
Try **A jacket costs $75. It has 20% off. GST of 9% is charged on the reduced cost.**

How much does Mei pay? $ ☐

Start here

Do these first. Take your time.

1 **The grid has 100 small squares. Some squares are shaded.**



(a) The percentage shaded is ☐ %.

(b) The percentage not shaded is ☐ %.

2 **Find each amount. Use 10%, 5% and 1%.**

(a) 10% of $340 = $ ☐

(b) 5% of $340 = $ ☐

(c) 15% of $340 = $ ☐

3 **Write each as a percentage.**

(a) 18 out of 25 pupils wear glasses. That is ☐ %.

(b) 7 out of 20 pupils cycle. That is ☐ %.

4 **A bag costs $120. It has 15% off.**

(a) 10% of $120 = $ ☐

(b) The discount is $ ☐

(c) The bag now costs $ ☐

Start here

Keep going. Check your work.

5 **A watch costs $64 before GST. GST is 9% of the cost before GST.**

(a) The GST is $ ☐

(b) The watch costs $ ☐ with GST.

6 **Ravi has $150. He spends 20% of his money on a game. Then he spends 40% of the money that is left on a book. How much money does Ravi have now? $**

☐

Working: _____

7 **There are 40 children in a group. 35% walk to school. 25% cycle. The rest take the bus.**

(a) ☐ children walk.

(b) ☐ children cycle.

(c) ☐ children take the bus.

8 **Mei scored 21 out of 30 in Quiz 1 and 34 out of 50 in Quiz 2.**

(a) Quiz 1 is ☐ %.

(b) Quiz 2 is ☐ %.

(c) She did better in Quiz 1 Quiz 2

9 **Caps cost $16 each. Shop P takes 15% off each cap. Shop Q sells the second cap at half cost. Siti buys 2 caps.**

(a) At Shop P she pays $ ☐

(b) At Shop Q she pays $ ☐

(c) The cheaper shop is Shop P Shop Q

Start here

Nearly there. You can do it.

10 **A shirt costs $50. The cost goes up by 20%. Then the new cost goes down by 20%.**

(a) After going up, the shirt costs $ ☐

(b) After going down, it costs $ ☐

(c) Is the cost back to $50? Yes No

11 **Each bar is cut into equal parts. Write the percentage that is shaded.**

(a) Bar 1: ☐ %

(b) Bar 2: ☐ %

Bar 1



Bar 2



12 **Circle True or False.**

(a) 50% of 80 is 40. True False

(b) 10% of 90 is 9. True False

(c) A 20% discount and then a 10% discount is the same as a 30% discount.
True False

13 **Nina has $240. She spends 30% of it on a bag. She spends 25% of the rest on a book. How much money does she have left? $** ☐

Working: _____

14 **Ali pays 9% GST on a $250 sofa. Then he pays $20 to have it delivered. The delivery has no GST. How much does he pay in all? $** ☐

Working: _____

Think harder

Hints are upside down. Try first!

1 **A $200 coat has 10% off. Then there is a further 20% off the new cost.**

(a) The coat finally costs $ ☐

(b) The 2 discounts together are ☐ % of $200.

Working: _____

_____

_____

_____

Hint. Take the 20% off the new cost, not off $200.

★ **Beyond Level 5: Working back from a percentage**

2 **After a 25% discount, a bag costs $60. What was the cost before the discount? $** ☐

Working: _____

_____

_____

_____

Hint. $60 is what percentage of the old cost?

★ **Beyond Level 5: GST in the cost**

3 **In this question GST is 8% of the cost before GST. A fridge costs $432 with GST.**

How much GST is in $432? $ ☐

Working: _____

_____

_____

_____

Hint. $432 is what percentage of the cost before GST?

★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 5: Percentage change

1 **A cost goes up from $60 to $75. Later it comes back down from $75 to $60. Work each percentage out from the amount that the change starts from.**

(a) The percentage increase from $60 to $75 is ☐ %.

(b) The percentage decrease from $75 to $60 is ☐ %.

Working: _____

_____

_____

Hint. Which amount does each change start from?

2 **A club has 50 members. 60% of them are girls. 30% of the girls and 20% of the boys wear a badge. What percentage of all the members wear a badge?**

☐ %

Working: _____

_____

_____

_____

Hint. Find the number of girls and boys first. Then count badges.

3 **A jar has 50 beads. 60% of the beads are blue. The rest are red. Blue beads are added to the jar, and no red beads are taken out. Now 80% of the beads are blue.**

How many blue beads were added? ☐

Working: _____

_____

_____

_____

Hint. Which beads do not change? Start with them.

✔ Check yourself

No hints. Try each one on your own.

1 **Find the amounts.**

(a) 10% of $180 = $ ☐ **(b)** 5% of $180 = $ ☐

(c) 15% of $180 = $ ☐

Working: _____

2 **A book costs $40. It has 25% off. GST of 9% is charged on the reduced cost. Find the total cost. $** ☐

Working: _____

3 **Kumar has $80. He spends 25% of it on lunch. He spends 50% of the rest on a gift. How much does he have left? $** ☐

Working: _____

4 **Test 1: 33 out of 50. Test 2: 16 out of 25. Circle the test with the higher percentage.** **Test 1** **Test 2**

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
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Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

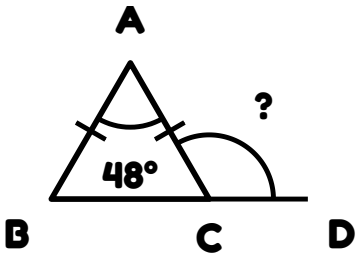
Learn it: Angles, triangles and quadrilaterals

My goal. I can use angle facts to find unknown angles step by step, and give a reason for every step.

Crossing lines make equal opposite angles. Drawings are not to scale.

Worked example

In the figure, AB = AC and angle BAC = 48°. BCD is a straight line. Find angle ACD. Give a reason for each step.



1. **AB = AC, so triangle ABC is isosceles. The angles at B and C are equal.**
2. **The angles of a triangle add up to 180°. 180° − 48° = 132°.**
3. **Angle ACB = 132° ÷ 2 = 66° (equal base angles).**
4. **BCD is a straight line, so ACD = 180° − 66° = 114° (angles on a straight line).**

Answer: Angle ACD = 114°.

How to

1. Find one angle. Write the reason.
2. Use it to find the next angle.
3. Check: do the angles add up?

Reasons

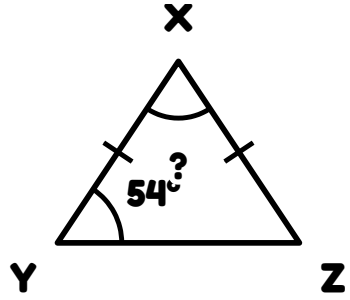
- Line 180°. Point 360°. Triangle 180°.
- Isosceles: equal sides, equal angles.
- Parallelogram: opposite angles equal, neighbours 180°.
- Trapezium: between parallel sides 180°.

Your turn

Answer at the back of the book.

Try

In triangle XYZ, XY = XZ and angle Y = 54°. Find angle X. ☐ °



Start here

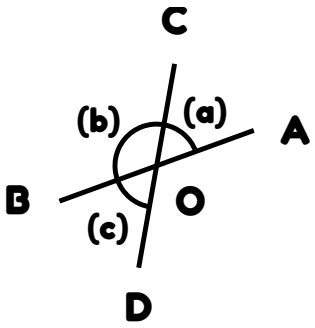
Do these first. Take your time.

1 **Lines AB and CD meet at O. Angle BOC is twice angle AOC. Find each angle.**

(a) Angle AOC (a) = ☐ °

(b) Angle BOC (b) = ☐ °

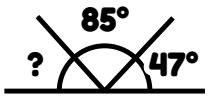
(c) Angle BOD (c) = ☐ °

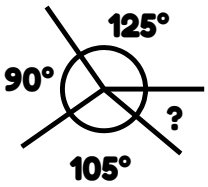


2 **Find the missing angles. The drawings are not to scale.**

(a) Straight line: ☐ °

(b) At the point: ☐ °

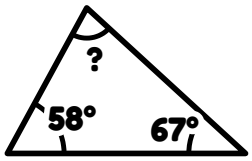


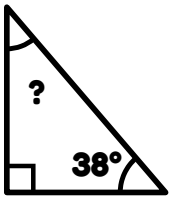


3 **Find the missing angle in each triangle. The second triangle has a right angle.**

(a) Left triangle: ☐ °

(b) Right triangle: ☐ °

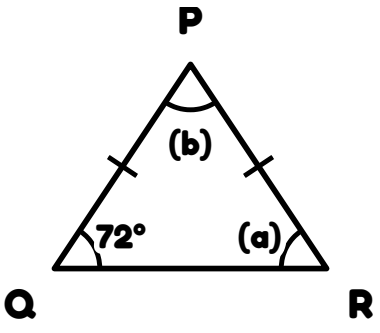




4 **In triangle PQR, PQ = PR. Angle Q = 72°.**

(a) Angle R (a) = ☐ °

(b) Angle P (b) = ☐ °



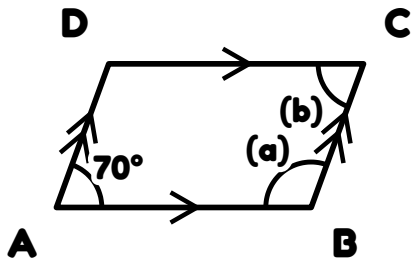
Start here

Keep going. Check your work.

5 **ABCD is a parallelogram. Angle A = 70°.**

(a) Angle B (a) = ☐ °

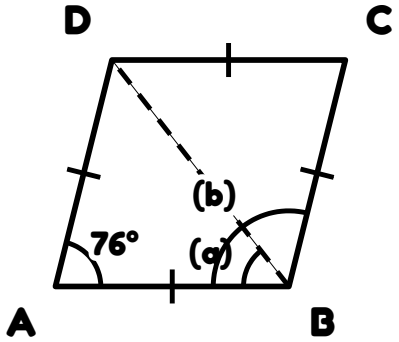
(b) Angle C (b) = ☐ °



6 **ABCD is a rhombus. Angle DAB = 76°. The dashed line is the diagonal BD.**

(a) Angle ABD (a) = ☐ °

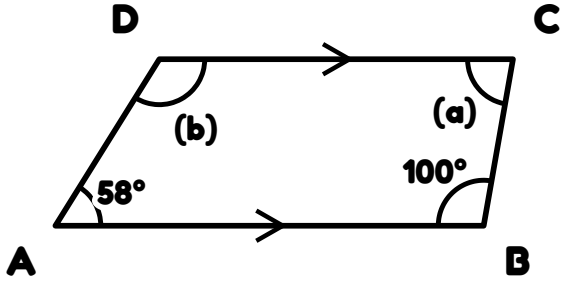
(b) Angle ABC (b) = ☐ °



7 **ABCD is a trapezium. AB is parallel to DC. Angle A = 58° and angle B = 100°.**

(a) Angle C (a) = ☐ °

(b) Angle D (b) = ☐ °



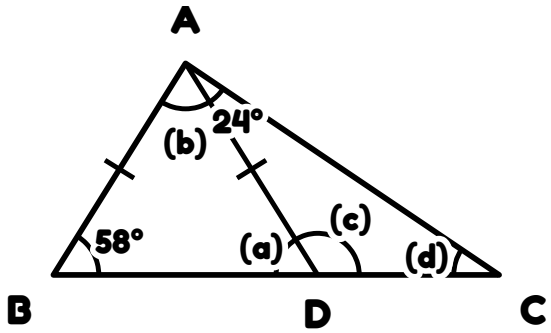
8 **BDC is a straight line. AB = AD. Angle ABD = 58° and angle DAC = 24°.**

(a) Angle ADB (a) = ☐ °

(b) Angle BAD (b) = ☐ °

(c) Angle ADC (c) = ☐ °

(d) Angle ACD (d) = ☐ °



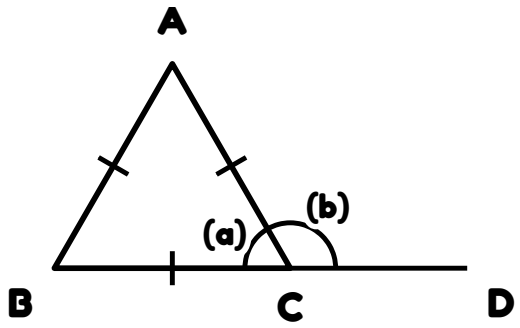
Start here

Nearly there. You can do it.

9 **ABC is an equilateral triangle. BCD is a straight line.**

(a) Angle ACB (a) = ☐ °

(b) Angle ACD (b) = ☐ °



10 **An isosceles triangle has an angle of 70°. The other 2 angles are not known yet.**

(a) If 70° is one of the 2 equal angles, the third angle is ☐ °.

(b) If 70° is the angle between the 2 equal sides, each equal angle is ☐ °.

(c) The number of different triangles is ☐.

11 **The minute hand of a clock turns through 360° in 60 minutes.**

(a) In 20 minutes it turns through ☐ °.

(b) At 4 o'clock the angle between the hands is ☐ °.

(c) At 7 o'clock the smaller angle between the hands is ☐ °.

12 **Circle the correct reason for each step.**

(a) Angle R = 50° because the
angles on a straight line | angles of a triangle | angles at a point
make 180°. (P = 55°, Q = 75°)

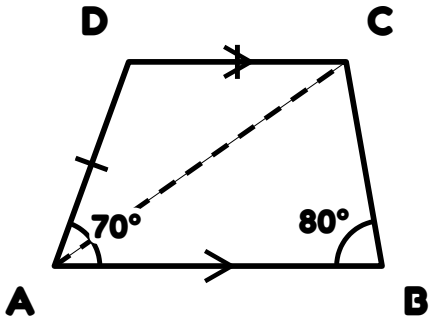
(b) 112° = 180° - 68° because
angles on a straight line | vertically opposite angles | angles at a point
make 180°.

(c) In a parallelogram A = 70° and B = 110° because
opposite angles | neighbouring angles | make 180°.

Think harder

Hints are upside down. Try first!

1 **ABCD is a trapezium. AB is parallel to DC. AD = DC. Angle A = 70° and angle B = 80°. The dashed line is the diagonal AC.**



(a) Angle ADC = ☐ °

(b) Angle DCA = ☐ °

(c) Angle ACB = ☐ °

Working: _____

Hint. Find angle D first.

2 **An isosceles triangle has whole-number angles. Find the angle at the top (between the 2 equal sides) in each case.**

(a) The top angle is 24° more than each equal angle. The top angle is ☐ °.

(b) Each equal angle is 24° more than the top angle. The top angle is ☐ °.

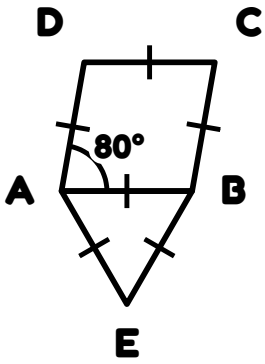
Working: _____

_____

Hint. Make 3 equal parts and the extra 24° (or 2 extra 24°s).

3

ABCD is a rhombus. ABE is an equilateral triangle. E is on the other side of AB from D. Angle DAB = 80°. Find angle AED. ☐ °



Working: _____

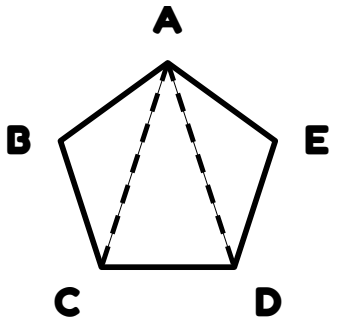
Hint. Which sides of the figure are equal in length?

★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 5: Angles of a regular pentagon

1 **The dashed lines split the pentagon into 3 triangles. A regular pentagon has 5 equal sides and 5 equal angles.**



(a) The 5 angles of the pentagon add up to ☐°.

(b) Each angle of a regular pentagon is ☐°.

Working: _____

Hint. Each triangle has angles that add up to 180°. Count the triangles.

★ Beyond Level 5: Clock angles

2 **The minute hand turns 6° each minute. The hour hand turns 0.5° each minute. At 4 o'clock the angle between the hands is 120°. Find the smaller angle between the hands at 4:20.** ☐°

Working: _____

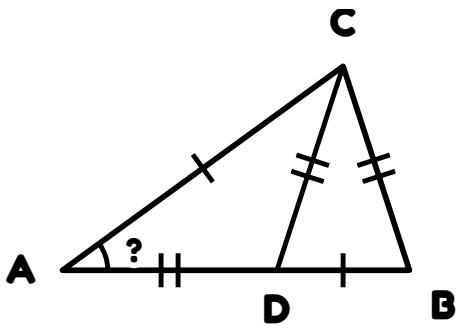
_____

Hint. At 4:20, how far has each hand turned from 12?

3

In triangle ABC, AB = AC. D is on AB so that AD = DC = CB. The marks show equal sides.

Find angle A. ☐°



Working: _____

_____

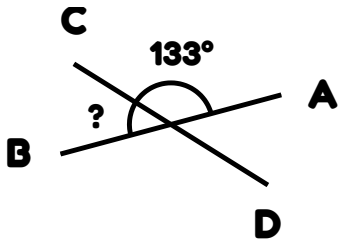
Hint. Call angle A one unit. Which triangles are isosceles?

✔ Check yourself

No hints. Try each one on your own.

1

Straight lines AB and CD meet at point O. Angle AOC = 133°. Find angle COB. ☐ °



2 **In triangle PQR, PQ = PR and angle P = 100°. Find angle Q.** ☐ °

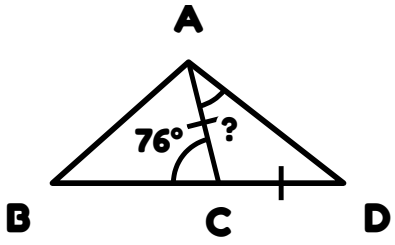
Working: _____

3 **ABCD is a parallelogram. Angle B = 115°. Find angle D.** ☐ °

Working: _____

4

BCD is a straight line. CA = CD. Angle ACB = 76°. Find angle CAD. ☐ °



My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

Learn it: Patterns and number sense

My goal. I can find the rule of a growing pattern and say it in words. I can use the rule for a far figure and backwards.

Find what grows each time. Make a table. Say the rule in words. Use it for any figure, and backwards.

Worked example

Tiles make a plus sign. Figure 1 has 5 tiles and figure 2 has 9 tiles. Figure 3 has arms of 3 tiles. How many tiles are in figure 10? Which figure has 81 tiles?

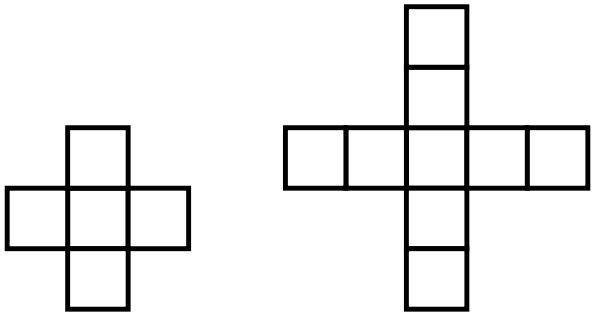


Figure 1 **Figure 2**

1. Each figure has 4 arms and 1 middle tile. In figure 3 each arm has 3 tiles.
2. Rule: tiles = 4 × figure number + 1. Figure 10: 4 × 10 + 1 = 41 tiles.
3. Backwards: 81 − 1 = 80, and 80 ÷ 4 = 20. Figure 20 has 81 tiles.

Answer: Figure 10 has 41 tiles. Figure 20 has 81 tiles.

How to

1. Count the figures. Make a table.
2. See what each new figure adds, and why.
3. Say the rule in words. Use it forwards and backwards.

Two kinds of thing

- Suppose all are the cheaper kind.
- Find how far short the total is.
- Divide by the difference in value.

Your turn

Answer at the back of the book.

Try **Matchstick hexagons are joined in a row. Each new hexagon shares one side with the hexagon before it. How many sticks do 12 hexagons need?**

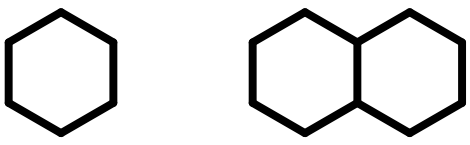


Figure 1 **Figure 2**

Start here

Do these first. Take your time.

1 **Matchstick hexagons are joined in a row. Each new hexagon shares one side with the hexagon before it. Figure 1 has 6 sticks. Figure 2 has 11 sticks.**

(a) 4 hexagons need ☐ sticks.

(b) 5 hexagons need ☐ sticks.

(c) Each new hexagon adds ☐ sticks.

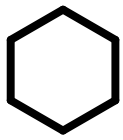


Figure 1

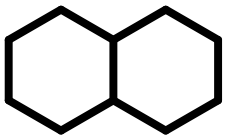


Figure 2

2 **Use the hexagon pattern. Complete the rule and use it.**

(a) sticks = ☐ × hexagons + ☐

(b) 35 hexagons need ☐ sticks.

3 **Mei has some matchsticks. She makes hexagons in a row, as in the pattern.**

(a) 96 sticks make ☐ hexagons.

(b) 100 sticks make at most ☐ hexagons, with ☐ sticks left over.

Working: _____

4 **Look at the pattern: 7, 12, 17, 22, ...**

(a) The next 2 numbers are ☐ and ☐.

(b) The 10th number is ☐.

(c) Is 100 in the pattern? **Yes** **No**

Working: _____

Start here

Keep going. Check your work.

5 **Tiles make U shapes. The bottom row of figure 1 has 3 tiles and a tile above each end. Each new figure has a bottom row 1 tile longer. Its 2 sides are 1 tile taller.**

(a) Figure 4 has ☐ tiles.

(b) Figure 15 has ☐ tiles.

(c) tiles = ☐ × figure + ☐

(d) Figure ☐ has 98 tiles.

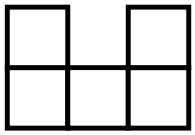


Figure 1

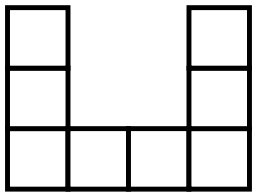


Figure 2

6 **Small badges cost $3 each and large badges cost $7 each. Dev buys 18 badges and pays $82 altogether.**

(a) Large badges: ☐

(b) Small badges: ☐

Working: _____

7 **A shop has 17 stools and chairs. A stool has 3 legs and a chair has 4 legs. There are 61 legs altogether.**

(a) There are ☐ chairs.

(b) There are ☐ stools.

Working: _____

8 **A quiz has 12 questions. A correct answer scores 5 marks. A wrong answer loses 2 marks. Siti answers every question and scores 39 marks.**

(a) She gets ☐ answers wrong.

(b) She gets ☐ answers right.

Working: _____

Start here

Nearly there. You can do it.

9 **Long tables are joined end to end in a row. Each table has 2 chairs on each long side. 1 chair sits at each end.**

(a) 5 tables have ☐ chairs.

(b) chairs = ☐ × tables + ☐

(c) 50 chairs belong to ☐ tables.

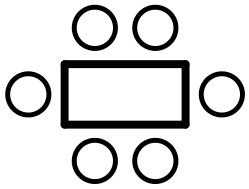


Figure 1

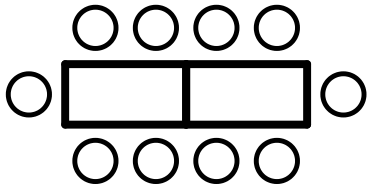


Figure 2

10 **The table shows a pattern.**

Figure	Sticks
1	7
2	10
3	13
4	16

(a) sticks = ☐ × figure number + ☐

(b) Figure 12 has ☐ sticks.

(c) The figure with 100 sticks is figure ☐.

11 **Circle True or False.**

(a) The numbers 5, 9, 13, 17, ... include 101. True False

(b) 10 hexagons need 51 sticks, so 20 hexagons need 102 sticks.
True False

(c) No number in the pattern 6, 11, 16, 21, ... ends in 3. True False

Think harder

Hints are upside down. Try first!

1 **A tin has 24 coins. Some are 10¢ coins and the rest are $1 coins. The coins are worth $7.80 altogether. How many $1 coins are there?** ☐

Working: _____

_____

Hint. Start by supposing all the coins are the same kind.

2 **Tiles make double staircases. Figure 1 has 2 tiles, figure 2 has 6 tiles, figure 3 has 12 tiles and figure 4 has 20 tiles.**

(a) Figure 10 has ☐ tiles.

(b) The figure with 156 tiles is figure ☐.

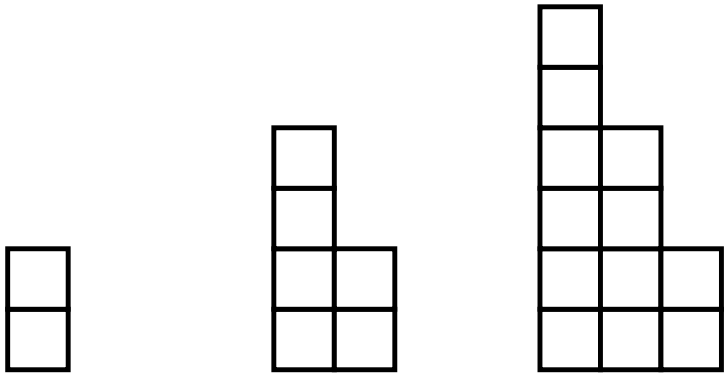


Figure 1 **Figure 2** **Figure 3**

Working: _____

Hint. Each figure adds a column 2 taller than the last. Use pairs.

3 **Look at the squares. Each new square adds an L-shaped layer of tiles (shaded). Find 1 + 3 + 5 + 7 + … + 49. The numbers go up by 2 each time.** ☐

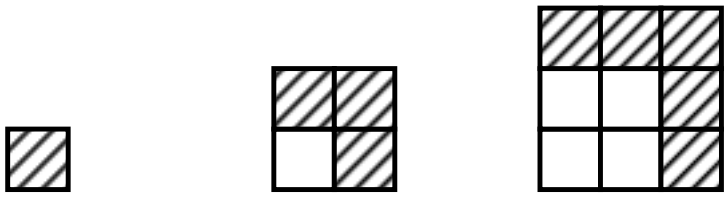


Figure 1 **Figure 2** **Figure 3**

Working: _____

Hint. 1, 1 + 3, 1 + 3 + 5: what kind of numbers are they?

★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 5: Adding a pattern

1 **Add 4 + 7 + 10 + 13 + … + 100. The numbers go up by 3 each time.** ☐

Working: _____

_____

_____

_____

Hint. How many numbers are there? Pair the first with the last.

★ Beyond Level 5: Handshakes

2 **Every person in a group shakes hands once with every other person.**

(a) 5 people make ☐ handshakes.

(b) 12 people make ☐ handshakes.

Working: _____

_____

_____

_____

Hint. Count each person's handshakes. Each one is counted twice.

3 **A test has 25 questions. A right answer scores 6 marks. A wrong answer loses 2 marks. A blank scores 0. Dev leaves 5 blank and scores 64 marks. How many questions does he get right?** ☐

Working: _____

_____

_____

_____

Hint. Only answered questions count. Suppose all of them are right.

✔ Check yourself

No hints. Try each one on your own.

1 **The numbers go 6, 11, 16, 21, … What is the 15th number?** ☐

Working: _____

2 **The rule for a pattern is: dots = 4 × figure number + 3. Which figure has 51 dots?** ☐

Working: _____

3 **A box holds 20 stools and chairs. A stool has 3 legs and a chair has 4 legs. There are 70 legs altogether. How many chairs are there?** ☐

Working: _____

4 **Matchstick hexagons are joined in a row. The rule is sticks = 5 × hexagons + 1. Mei has 130 sticks. What is the most hexagons she can make?** ☐

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

Learn it: Mixed heuristics

My goal. I can choose a way to start a puzzle, and check my answer against every clue.

Choose a start: draw a model, make a list or a table, work backwards, guess and check, or use a logic grid. Then check your answer against every clue.

Worked example

Ali, Mei and Siti each have a different pet: a dog, a cat or a fish. Ali does not have a fish. Mei does not have a cat or a fish. Who has the cat?

	Dog	Cat	Fish
Ali	No	Yes	No
Mei	Yes	No	No
Siti	No	No	Yes

1. Mei does not have a cat or a fish, so Mei has the dog. Write Yes for Mei and dog.
2. The dog is taken. Write No in the dog column for Ali and Siti, and No in the other columns for Mei.
3. Ali has no fish and no dog, so Ali has the cat.
4. Check: Siti has the fish. Every clue is true.

Answer: Ali has the cat.

How to

1. Use a clue that tells you something for sure.
2. A Yes means No in the rest of its row and column.
3. Check every clue at the end.

Ways to start

- Draw a model.
- List in order or make a table.
- Work backwards.
- Guess and check.

Your turn

Answer at the back of the book.

Try **Ben, Kumar and Aisha each choose a different drink: tea, juice or milk. Ben does not choose juice or milk. Aisha does not choose tea. Kumar does not choose milk. Who chooses juice?** ☐

Name _____

Start here

Do these first. Take your time.

1 **Mei picks a number. She multiplies it by 3, adds 8 and then halves the result. She gets 19. What number did Mei pick?** ☐

Working: _____

_____

2 **Kumar spends 1/4 of his money. Then he spends $6. He has $15 left. How much money did Kumar have at first? $** ☐

Working: _____

3 **Ali, Mei and Siti ran a race. Ali did not win. Siti was not last. Mei finished after Siti. Who won the race?** ☐

4 **Mei and Kumar have $75 together. Kumar has $19 more than Mei.**

(a) Kumar has $ ☐ .

(b) Mei has $ ☐ .

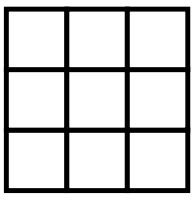
5 **Count the squares of every size in a 3 by 3 grid.**

(a) Squares 1 by 1: ☐

(b) Squares 2 by 2: ☐

(c) Squares 3 by 3: ☐

(d) All the squares: ☐



Start here

Keep going. Check your work.

6 **One of 3 children ate the last slice of cake. Exactly one of the 3 children is telling the truth.**
Kai says, "Mo ate it."
Lin says, "I did not eat it."
Mo says, "Kai is lying." Who ate the slice? ☐

Working: _____

7 **Mum is 36 years old. Dev is 8 years old. In how many years will Dev be half as old as Mum?** ☐ **years**

Working: _____

8 **Ben gives 1/3 of his stickers to Ali. He gives 1/2 of the rest to Mei. 12 stickers are left. Find the number of stickers Ben began with.** ☐

Working: _____

9 **A farm has hens and cows, 20 animals in all. They have 56 legs altogether. Hens have 2 legs each and cows have 4 legs each.**

(a) There are ☐ cows.

(b) There are ☐ hens.

Working: _____

Start here

Nearly there. You can do it.

10 **The product of 2 whole numbers is 72. Their sum is 17. List the pairs with product 72 to find them.**

(a) The numbers are ☐ and ☐.

Working: _____

11 **Use the digit cards 1, 2 and 3 once each to make 3-digit numbers.**



(a) The number of different numbers is ☐.

(b) The even numbers are ☐ and ☐.

(c) The number closest to 250 is ☐.

Working: _____

12 **Ali writes the whole numbers from 1 to 80. How many times does he write the digit 7?** ☐

Working: _____

Think harder

Hints are upside down. Try first!

1 **Ben, Kumar, Aisha and Wei each play a different sport: swimming, tennis, football or badminton. Write Yes or No in the grid. Then write the sport for each child.**
Clue 1: Ben does not play tennis or badminton.
Clue 2: Kumar does not play swimming or tennis.
Clue 3: Wei does not play swimming, badminton or tennis.

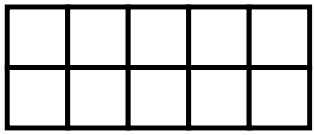
	swimming	tennis	football	badminton
Ben				
Kumar				
Aisha				
Wei				

(a) Ben plays ☐ **(b)** Kumar plays ☐

(c) Aisha plays ☐ **(d)** Wei plays ☐

Hint. Which child has the most clues? Start with Wei.

2 **Count the squares and the rectangles in a 2 by 5 grid. Every square is also a rectangle, so count it in both.**



(a) The number of squares (all sizes) is ☐.

(b) The number of rectangles (all sizes) is ☐.

Working: _____

Hint. Count by shape: first the ones 1 high, then the ones 2 high.

3 **In a cafe, 1/4 of the customers leave. Then 3 more customers arrive. Then 1/3 of the customers in the cafe leave. 14 customers are left. How many customers were in the cafe at first?** ☐

Working: _____

Hint. Undo the last step first. 14 is what part of the number before?

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 **How many 3-digit numbers have 3 different digits that add up to 8?** ☐

Working: _____

_____

_____

Hint. Find the sets of 3 digits first. Then count the orders of each set.

2 **Dev has 30 coins. They are 5¢, 10¢ and 20¢ coins. The number of 5¢ coins equals the number of 20¢ coins. The coins are worth $3.45 in all.**

(a) There are ☐ coins of 5¢.

(b) There are ☐ coins of 20¢.

(c) There are ☐ coins of 10¢.

Working: _____

_____

_____

Hint. A 5¢ coin and a 20¢ coin come as a pair. What is a pair worth?

★ **Beyond Level 5: The worst case**

3 **A drawer has 10 red socks, 10 blue socks and 10 black socks. You take socks out in the dark. How many socks must you take to be sure of 2 socks of the same colour?** ☐

Working: _____

_____

_____

Hint. Think of the worst luck. How many socks can you take with no pair?

✔ Check yourself

No hints. Try each one on your own.

1 **Siti picks a number. She adds 7 and then multiplies by 4. She gets 60. What number did Siti pick?** ☐

Working: _____

2 **Mei spends 2/5 of her money. Then she spends $9. She has $12 left. How much money did Mei have at first? $** ☐

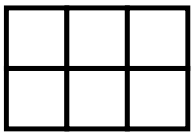
Working: _____

3 **A box holds 9 hens and goats. They have 26 legs altogether. A hen has 2 legs and a goat has 4 legs. How many goats are there?** ☐

Working: _____

4

Count the squares of every size in a 2 by 3 grid. ☐



My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

My progress

Tick each box when you finish. Write your Check yourself score.

Unit	Topic	Learn it	Start here	Think harder	Go further	Score	Date
1	Big number puzzles	☐	☐	☐	☐	/ 4	
2	Fractions at work	☐	☐	☐	☐	/ 4	
3	Change and compare	☐	☐	☐	☐	/ 4	
4	Half the box	☐	☐	☐	☐	/ 4	
5	Fill it up	☐	☐	☐	☐	/ 4	
6	Rates and deals	☐	☐	☐	☐	/ 4	
7	Out of 100	☐	☐	☐	☐	/ 4	
8	Chase the angle	☐	☐	☐	☐	/ 4	
9	Rules for patterns	☐	☐	☐	☐	/ 4	
10	Pick a start	☐	☐	☐	☐	/ 4	

Look back

My favourite unit was _____

The hardest idea was _____

Next I want to try _____



Certificate of Completion

Well done,

You finished the Math Inquirers Extension Workbook, Level 5.

Date

Signed by a grown-up

Answers

Try each question first. Then check your work here.

Each answer says **Answer** first, then **Why**, and the page of the question. Got one wrong? Read the Why, then try it again. Stuck on the Why? Ask someone to read it with you.

Cover this part of the book with a folded sheet of paper while you work, so you do not see the answers by mistake.

Unit 1 · Big number puzzles

Your turn
page 6

Answer: 7 964 310
Why: Just below 8 000 000: start with 7, then the greatest order: 7 964 310 (35 690 away). Just above: start with 9, then the smallest order: 9 013 467 (1 013 467 away). 7 964 310 is closer.

START HERE

1
page 7

Answer: (a) 9 040 007 (b) 70 000 (c) 53 000
Why: (a) 9 million, 040 thousand, 007: 9 040 007. (b) The 7 is in the ten thousands place: 70 000. (c) 5 300 000 ÷ 100 = 53 000.

2
page 7

Answer: 940 621, 4 096 213, 4 096 231, 4 906 213
Why: 940 621 has 6 digits, so it is smallest. Then compare 4 096 213, 4 096 231 and 4 906 213 digit by digit.

3
page 7

Answer: (a) 7 850 000 (b) 7 800 000 (c) 8 000 000
Why: Look at the digit after the place. (a) 849 520: the hundreds digit is 5, so 7 850 000. (b) 49 520 is under 50 000, so 7 800 000. (c) 849 520 is over 500 000, so 8 000 000.

4
page 7

Answer: (a) 9 875 410 (b) 1 045 789
Why: (a) A multiple of 5 ends in 0 or 5. Ending in 0: 9 875 410. Ending in 5 gives only 9 874 105. (b) Start with 1, then 0, 4, 5, 7, 8, 9: 1 045 789.

5
page 7

Answer: 5 971 946
Why: 3 hundred thousands is 300 000, and 6 ten thousands is 60 000. 5 731 946 + 300 000 = 6 031 946. Then 6 031 946 - 60 000 = 5 971 946.

6
page 8

Answer: (a) must be wrong (b) could be right (c) could be right (d) must be wrong
Why: (a) 4000 × 50 = 200 000, but 20 722 is far too small. (b) 4 million + 3 million is about 7 million: close. (c) 8000 × 300 = 2 400 000: close. (d) 6000 ÷ 15 = 400, so 4050 is 10 times too big.

7
page 8

Answer: (a) (9 - 2) × 3 + 1 = 22 (b) 9 - 2 × (3 + 1) = 1 (c) 2 400 000 ÷ (300 + 100) × 2 = 12 000
Why: (a) (9 - 2) × 3 + 1 = 22. (b) 9 - 2 × (3 + 1) = 1. (c) 300 + 100 = 400, 2 400 000 ÷ 400 = 6000, × 2 = 12 000.

8
page 8

Answer: 104 cartons
Why: 1 250 000 ÷ 500 = 2500 boxes. Then 2500 ÷ 24 = 104 remainder 4, since 24 × 104 = 2496. Only full cartons count, so 104.

9
page 8

Answer: (a) 8650 (b) 8749
Why: The halfway marks are 8650 and 8750. 8650 rounds up to 8700, so it counts. 8750 rounds up to 8800, so the greatest is 8749.

10
page 8

Answer: (a) True (b) False (c) True
Why: (a) 54 × 100 000 = 5 400 000: true. (b) 1 000 000 - 1 = 999 999: false. (c) 999 999 is 1 less than 1 000 000: true.

11 page 9
Answer: (a) Dune (b) 395 550
Why: (a) Compare the hundred thousands: Dune has 4. (b) Cedar is smallest: 1 420 050 - 1 024 500 = 395 550.

12 page 9
Answer: 472 630
Why: Clue 2: 465 000 to 474 999. Clue 3: thousands digit 2, so 472 (462 is too low). Clue 5: ones 0. Clue 4: tens and hundreds are 3 and 6 (1 and 2 or 4 and 8 repeat a digit). So 472 630.

13 page 9
Answer: 14 250 people
Why: Monday 3100, Tuesday 6200, Wednesday 4950. Total 14 250. The 48 000 is not needed.

14 page 9
Answer: 4 500 000 and 5 499 999
Why: Numbers from 4 500 000 to 5 499 999 round to 5 000 000.

THINK HARDER

1 page 10
Answer: 6 435 708
Why: Clue 2: 6 350 000 to 6 449 999. Clue 3 leaves 6 4 3, and the tens digit is 0. The other 3 digits add to 33 - 6 - 4 - 3 = 20. With thousands t and hundreds t + 2, only t = 5 keeps all digits different: 5, 7 and ones 8. So 6 435 708.

2 page 10
Answer: (a) 10
Why: Below 200 000 the digits 1 + 9 are already 10. So the number is 20 _ _ _ _ with the other digits adding to 2: 4 ways with one 2 and 6 ways with two 1s. Total 10.

3 page 10
Answer: 69 936
Why: Put the greatest digits first: 9 and 7. 752 × 93 = 69 936 beats 953 × 72 = 68 616 and 732 × 95 = 69 540.

GO FURTHER

1 page 11
Answer: 544 320
Why: First digit: 9 choices (1 to 9). Second: 9 choices (0 is allowed). Then 8, 7, 6, 5, 4. 9 × 9 × 8 × 7 × 6 × 5 × 4 = 544 320.

2 page 11
Answer: 13 500
Why: Write 7 as 007. The numbers 000 to 999 are 1000 numbers. In each of the 3 places every digit 0 to 9 appears 100 times, adding 4500. 3 × 4500 = 13 500.

3 page 11
Answer: 101 (3 × 4 × 9 - 7)
Why: 3 × 4 × 9 - 7 = 101, which is 1 away. 99 and 100 cannot be made, and 102 is 2 away. So 101.

4 page 11
Answer: 53 days
Why: 2400 × 8 = 19 200 sheets a day. 52 days give 998 400, not enough, so 53 days.

CHECK YOURSELF

1 page 12
Answer: (a) 200 000 (b) 5 300 000
Why: (a) The 2 is in the hundred thousands place: 200 000. (b) 5 280 436: the ten thousands digit is 8, so round up to 5 300 000.

2 page 12
Answer: 3 976 520
Why: Just below: 3 976 520 (23 480 away). Just above: 5 023 679 (1 023 679 away). The first is closer.

3 page 12
Answer: (8 + 12) ÷ 4 × 2 = 10
Why: (8 + 12) ÷ 4 × 2 = 20 ÷ 4 × 2 = 5 × 2 = 10. Without brackets the line gives 8 + 3 × 2 = 14.

4 page 12
Answer: 270 000 passengers
Why: In 1 week: 18 × 1250 = 22 500. In 12 weeks: 22 500 × 12 = 270 000.

Unit 2 · Fractions at work

Your turn page 13
Answer: 30 beads
Why: 1/4 of 120 = 30, so 90 beads are left. 2/3 of 90 = 60 beads for the necklace. 90 - 60 = 30 beads are left.

START HERE

1 page 14
Answer: (a) 5/6 (b) 3/8 (c) 2 1/4
Why: a ÷ b = a/b. (a) 5/6. (b) 3/8. (c) 9 ÷ 4 = 2 and 1/4.

2 page 14
Answer: (a) 5/6 (b) 4/7 (c) 2/3 of 24
Why: (a) 1/6 is less than 1/5 (shortfall from 1), so 5/6. (b) 35/63 and 36/63: 4/7. (c) 15 and 16: 2/3 of 24.

3 page 14
Answer: (a) 24 (b) 40 (c) 16
Why: 1 part is 64 ÷ 8 = 8 cm. (a) 3 parts = 24 cm. (b) 5 parts = 40 cm. (c) 1/4 is 2 parts of the 8, which is 16 cm.

4 page 14
Answer: (a) 11/12 (b) 11/24 (c) 7/12
Why: Change to the same bottom number. (a) 8/12 + 3/12 = 11/12. (b) 20/24 - 9/24 = 11/24. (c) 9/12 - 2/12 = 7/12.

5 page 15
Answer: 4 1/2 km
Why: 6 × 3/4 = 18/4. Divide 18 by 4: 4 with 2 left over, so 4 and 2/4, which is 4 1/2. Or: 3/4 of 6 is 4 1/2.

6 page 15
Answer: 15 cards
Why: 3/8 of 72 = 27, so 72 - 27 = 45 are left. 2/3 of 45 = 30 go to Ben. 45 - 30 = 15.

7 page 15
Answer: 1/4 (7/12 = 1/3 + 1/4)
Why: 1/3 = 4/12. 7/12 - 4/12 = 3/12 = 1/4. Both 1/3 and 1/4 have 1 on top and they are different, so 7/12 = 1/3 + 1/4.

8 page 15
Answer: (a) 5/11, 6/13 (b) 10/21 (c) 13
Why: The top is the position and the bottom is 2 × position + 1. (a) 5/11 and 6/13. (b) 10/21. (c) 13/27: 2 × 13 + 1 = 27.

9 page 15
Answer: 7/20 of the book
Why: 2/5 + 1/4 = 8/20 + 5/20 = 13/20 is read. The whole book is 20/20. 20/20 - 13/20 = 7/20.

10 page 15
Answer: (a) 1 1/4 (b) 1 3/4
Why: Each mark is 1/4. A is 1 and 1 quarter: 1 1/4. B is 1 3/4.

11 page 16
Answer: (a) 1/4 (b) 2/5 (c) 1 1/3
Why: Multiply the tops and the bottoms, then simplify. (a) 6/24 = 1/4. (b) 6/15 = 2/5. (c) 12/9 = 4/3 = 1 1/3.

12 page 16
Answer: (a) 18 (b) 54
Why: The book is 1 part, lunch is 1 part and the $18 left is 1 part. So 3 parts = $54.

13 page 16
Answer: 15 children
Why: 3/8 of 40 = 15 walk. 1/4 of 40 = 10 take the bus. 15 + 10 = 25, so 40 - 25 = 15 cycle.

14 page 16
Answer: (a) False (b) True (c) True
Why: (a) 3 ÷ 5 = 3/5, not 5/3. (b) 7/8 = 21/24 and 5/6 = 20/24, so it is true. (c) 1/4 × 1/2 = 1/8, so it is true.

THINK HARDER

1 page 17
Answer: 1/3 + 1/15
Why: 2/5 - 1/3 = 1/15: a unit fraction. 2/5 - 1/4 = 3/20: no. No other unit fraction is between 1/5 and 2/5, so 1/3 + 1/15.

2 page 17
Answer: 120 eggs
Why: After the morning 2/3 are left. Then 3/4 of 2/3 = 1/2. Then 3/5 of 1/2 = 3/10. 36 is 3/10, so 1/10 is 12 and the whole is 120.

3 page 17
Answer: (a) 5 1/6 (b) 29/35
Why: Greatest: big top, small bottom: 7/2 + 5/3 = 31/6 = 5 1/6. Smallest: small tops, big bottoms: 2/5 + 3/7 = 29/35.

GO FURTHER

1 page 18
Answer: 1/12 of the cake
Why: Cut every sixth into 2 equal parts. Then 5/6 becomes 10 parts, and each part is 1/12 of the cake. So 5/6 ÷ 10 = 1/12.

2 page 18
Answer: 4 ways
Why: Test 1/7, 1/8, 1/9 and 1/10: each leaves a unit fraction (1/42, 1/24, 1/18, 1/15). 1/11 leaves 5/66: no. 1/12 is half of 1/6, so stop. 4 ways.

3 page 18
Answer: (a) 64 (b) 21
Why: Work backwards: 27 ÷ 3/4 = 36, 36 ÷ 3/4 = 48, 48 ÷ 3/4 = 64. Heights 64, 48, 36, 27. (b) 48 - 27 = 21.

4 page 18
Answer: 4/9
Why: Write each as 1/2 × (a difference). The middle parts cancel, leaving 1/2 × (1 - 1/9) = 4/9.

CHECK YOURSELF

1 page 19
Answer: (a) 4/9 (b) 2 1/3
Why: (a) 4 ÷ 9 = 4/9. (b) 7 ÷ 3: 3 goes into 7 two times with 1 left, so 2 and 1/3.

2 page 19
Answer: (a) 13/24 (b) 7/10
Why: (a) 21/24 - 8/24 = 13/24. (b) 4/10 + 3/10 = 7/10.

3 page 19
Answer: 60 marbles
Why: 1/5 of 120 = 24, so 96 are left. 3/8 of 96 = 36 go to Wei. 96 - 36 = 60.

4 page 19
Answer: 1/12 of the bag
Why: 3/4 + 1/6 = 9/12 + 2/12 = 11/12 is used. 12/12 - 11/12 = 1/12 is left.

Unit 3 · Change and compare

Your turn page 20
Answer: 25 stickers
Why: The total stays the same: 70 + 50 = 120. After, Jason has 3 units and Amelia has 5 units. 8 units = 120, so 1 unit = 15. Jason now has 3 × 15 = 45. Jason gives 70 - 45 = 25 stickers.

START HERE

1 page 21
Answer: (a) 36 (b) 24 (c) 36
Why: (a) 3/8 of 96 = 36. (b) 96 - 36 = 60 is left. 2/5 of 60 = 24. (c) 60 - 24 = 36 is left.

2 page 21
Answer: (a) 17 (b) 19 (c) 36
Why: (a) 22 - 5 = 17. (b) 14 + 5 = 19. (c) 22 + 14 = 36, and 17 + 19 = 36. Giving does not change the total.

3 page 21
Answer: (a) 17, 46 (b) 29, 29
Why: (a) 17 and 46. (b) 29 and 29: equal years added keep the difference.

4 page 21
Answer: 9 books
Why: The total stays 52 + 34 = 86. Each shelf has 43. The first shelf loses 52 - 43 = 9 books.

5 page 21
Answer: 12 sweets
Why: 1/12 of 72 = 6. 5/12 of 72 = 30 are mints. 72 - 30 = 42 are not. 42 - 30 = 12 more.

6 page 22
Answer: (a) blue (b) 2/5 (c) 4/15
Why: (a) Only red cubes were added. (b) 8 of 20 = 2/5. (c) 8 of 30 = 4/15.

7 page 22
Answer: (a) 15 (b) 75
Why: The bag takes 2 of the 5 parts. The other 3 parts are $45, so 1 part = 45 ÷ 3 = 15. All 5 parts = 5 × 15 = 75.

8 page 22
Answer: (a) 21 (b) 7
Why: The book is 3 units and the pen is 1 unit. 4 units = 28, so 1 unit = 7. The pen costs 7. The book costs 3 × 7 = 21.

9 page 22
Answer: (a) 32 (b) 24
Why: Siti has 4 units and Mei has 3 units. 7 units = 56, so 1 unit = 8. Siti has 4 × 8 = 32. Mei has 3 × 8 = 24.

10 page 23
Answer: (a) the total (b) the difference (c) the blue beads
Why: (a) Ben gains what Ali loses: the total. (b) Both gain $5: the difference. (c) Only red beads go in: the blue beads.

11 page 23
Answer: (a) 68 (b) 41 (c) 27
Why: (a) 56 + 12 = 68. (b) 29 + 12 = 41. (c) 68 - 41 = 27. Before the gift 56 - 29 = 27 too. The difference did not change.

12 page 23
Answer: $80
Why: Shirt 2 units, shoes 3 units, left 5 units of 10 = $40. 1 unit = $8, so $80.

13 page 23
Answer: 750 ml
Why: 5/6 - 1/2 = 5/6 - 3/6 = 2/6 = 1/3 of the jug was poured out. 1/3 is 250 ml, so a full jug is 3 × 250 = 750 ml.

14 page 23
Answer: (a) False (b) True (c) True
Why: (a) False: what Ali loses Ben gains. (b) True: both amounts go up by 5. (c) True: if Mei has 1 unit, Ali has 2 units.

THINK HARDER

1 page 24
Answer: $160
Why: Monday leaves 5/8. Tuesday takes 2/5 of 5/8 = 2/8. Left 3/8 = $60, so 1/8 = $20 and the whole is $160.

2 page 24
Answer: 62 cards
Why: After: Ben 4 units, Ali 7 units. 11 units = 132, so 1 unit = 12. Ben has 48 now. Before giving 14 he had 48 + 14 = 62.

3 page 24
Answer: 3 years
Why: The difference stays 30 years. The son is 1 unit and the father 3 units, so 2 units = 30. 1 unit = 15, which is 3 years from now.

GO FURTHER

1 page 25
Answer: (a) 30 (b) 40 (c) 80
Why: Cara 1 unit, Ben 1 unit + 10, Ali 2 units + 20. 4 units + 30 = 150, so 1 unit = 30. Cara 30, Ben 40, Ali 80.

2 page 25
Answer: 150 people
Why: The total is the same. 1/2 - 2/5 = 1/10 of the total is 15, so the total is 150.

3 page 25
Answer: 168 beads
Why: The white beads stay the same: 5/7 of the start and 5/6 of the end. White is 5 parts, so the total goes from 7 parts to 6 parts. 1 part = 24, so 7 × 24 = 168.

4 page 25
Answer: 45 marbles
Why: Ali has 3 units and keeps 2. Ben then has 6 units. 2 + 6 = 8 units = 120, so 1 unit = 15 and Ali had 45.

CHECK YOURSELF

1 page 26
Answer: $60
Why: Book 1 unit, bag 2 units, left 2 units = $24. 1 unit = $12, so $60.

2 page 26
Answer: 8 cards
Why: The total is 76, so each has 38. Jamie has 46 - 38 = 8 too many. Jamie gives 8 cards.

3 page 26
Answer: $30
Why: The difference stays $20 = 1 unit. Amelia now has $80, so the gift is $30.

4 page 26
Answer: (a) ducks (b) 3/8
Why: (a) Only geese came, so the 24 ducks did not change. (b) Now there are 24 + 40 = 64 birds. 24/64 = 3/8.

Unit 4 · Half the box

Your turn page 27
Answer: 54 cm²
Why: The right angle is at Q, so the sides PQ and QR meet at a right angle. Take QR as the base and PQ as the height. 1/2 × 12 × 9 = 54. The side PR = 15 cm is slanting, so it is not used.

START HERE

1 page 28
Answer: (a) 12 (b) 84
Why: (a) 1/2 × 8 × 3 = 12. (b) 1/2 × 14 × 12 = 84. Slanting sides are not used.

2 page 28
Answer: (a) 24 (b) 6 (c) QR
Why: (a) 1/2 × 8 × 6 = 24. (b) PQ meets QR at Q: 6 cm. (c) With base PQ the height is QR.

3 page 28
Answer: (a) 8 (b) 10
Why: Base × height is 2 times the area. (a) 2 × 36 = 72 and 72 ÷ 9 = 8. (b) 2 × 45 = 90 and 90 ÷ 9 = 10.

4 page 28
Answer: (a) 42 (b) is the same
Why: (a) The base is 12 cm and the height is 7 cm: 1/2 × 12 × 7 = 42. (b) Wherever P is, the base and height stay the same.

5 page 29
Answer: 84 cm²
Why: The square is 8 × 8 = 64 cm². The triangle is 1/2 × 8 × 5 = 20 cm². Add: 64 + 20 = 84 cm².

6 page 29
Answer: (a) 63 (b) 63
Why: (a) 1/2 × 14 × 9 = 63. (b) The rectangle is 14 × 9 = 126 cm². 126 - 63 = 63. The 2 white triangles also add to half of the rectangle.

7 page 29
Answer: 15 cm²
Why: The base AB is 6 squares long. The top point C is 5 squares above the base, so the height is 5. 1/2 × 6 × 5 = 15 cm².

8 page 29
Answer: 66 cm²
Why: Use BC as the base. The height AH is 12 cm. 1/2 × 11 × 12 = 66. The sides 20 and 13 are not used.

9 page 30
Answer: (a) 144 (b) 16
Why: (a) 12 × 12 = 144. (b) The triangle has area 144 too, so base × height = 288. 288 ÷ 18 = 16.

10 page 30
Answer: 98 cm²
Why: The full rectangle is 12 × 9 = 108. The cut corner has legs 4 and 5: area 10. 108 - 10 = 98 cm².

11 page 30
Answer: (a) 6 (b) 6
Why: Base × height = 18. The pairs are 1 and 18, 2 and 9, 3 and 6, 6 and 3, 9 and 2, 18 and 1: 6 pairs. A base of 3 goes with a height of 6.

12 page 30
Answer: (a) True (b) False (c) True
Why: (a) True. (b) False: a slanting side is not at a right angle. (c) True: both are 30 cm².

THINK HARDER

1 page 31
Answer: 4.8 cm
Why: 1/2 × 8 × 6 = 24. With base 10: 1/2 × 10 × height = 24, so the height is 4.8 cm.

2 page 31
Answer: 52.5 cm²
Why: Rectangle 14 × 10 = 140. White triangles: 35 + 17.5 + 35 = 87.5. 140 - 87.5 = 52.5 cm².

3 page 31
Answer: (a) 24 (b) 12
Why: (a) Same height, and BD is 1/3 of BC: 1/3 of 72 = 24. (b) Same height from B, and DE is half of DA: half of 24 = 12.

GO FURTHER

1 page 32
Answer: 112 cm²
Why: Square 16 × 16 = 256. White triangles: 32 + 48 + 64 = 144. 256 - 144 = 112 cm².

2 page 32
Answer: 13 cm²
Why: The rectangle round it is 6 × 5 = 30. The 3 outside triangles are 6, 6 and 5: 17. 30 - 17 = 13 cm².

3 page 32
Answer: 8 cm²
Why: Draw DC. ADC = 1/4 of 96 = 24. ADE = 1/3 of ADC = 8 cm².

CHECK YOURSELF

1 page 33
Answer: 48 cm²
Why: 1/2 × 16 × 6 = 48. The sides 10 and 10 are not used.

2 page 33
Answer: 10 cm
Why: 2 × 70 = 140 is base × height. 140 ÷ 14 = 10.

3 page 33
Answer: 72 cm²
Why: Rectangle 10 × 6 = 60. The triangle has base 6 and height 14 - 10 = 4: area 12. Total 72 cm².

4 page 33
Answer: 15 bags
Why: The area is 1/2 × 18 × 10 = 90 m². 90 ÷ 6 = 15 bags.

Unit 5 · Fill it up

Your turn page 34
Answer: 13 cm
Why: The base area is 25 × 20 = 500 cm². The stone raises the water by 1500 ÷ 500 = 3 cm. The new depth is 10 + 3 = 13 cm.

START HERE

1 page 35
Answer: (a) 5 (b) 10 (c) 30, 30
Why: (a) 5 long. (b) 5 × 2 = 10. (c) 10 × 3 = 30 cubes, so 30 cm³.

2 page 35
Answer: (a) 270 (b) 343
Why: (a) 9 × 6 × 5 = 54 × 5 = 270 cm³. (b) 7 × 7 × 7 = 49 × 7 = 343 cm³.

3 page 35
Answer: (a) 5 (b) 6
Why: (a) The base area is 8 × 5 = 40 cm². 200 ÷ 40 = 5 cm. (b) The base area is 144 ÷ 4 = 36 cm². 6 × 6 = 36, so each side is 6 cm.

4 page 35
Answer: (a) 3500 (b) 12 (c) 60
Why: (a) 3.5 × 1000 = 3500. (b) 12 000 ÷ 1000 = 12. (c) 50 × 40 × 30 = 60 000 cm³, and 60 000 ÷ 1000 = 60 litres.

5 page 36
Answer: (a) 14400 (b) 14.4 (c) 15.6
Why: (a) 40 × 30 × 12 = 14 400 cm³. (b) 14.4 litres. (c) Full: 30 000 cm³ = 30 litres. 30 - 14.4 = 15.6.

6 page 36
Answer: (a) 5 (b) 12
Why: The base area is 40 × 25 = 1000 cm². 5 litres = 5000 cm³. The rise is 5000 ÷ 1000 = 5 cm. The new depth is 7 + 5 = 12 cm.

7 page 36
Answer: 1080 cm³
Why: The rise is 11 - 8 = 3 cm. The volume of the stone is 24 × 15 × 3 = 360 × 3 = 1080 cm³.

8 page 36
Answer: (a) 5 (b) 15
Why: The base area is 60 × 30 = 1800 cm². 9 litres = 9000 cm³. The fall is 9000 ÷ 1800 = 5 cm. The new depth is 20 - 5 = 15 cm.

9 page 36
Answer: (a) 5 (b) 10
Why: Edges with product 30: 1, 1, 30; 1, 2, 15; 1, 3, 10; 1, 5, 6; 2, 3, 5. That is 5. The sums are 32, 18, 14, 12, 10: smallest 10.

10 page 37
Answer: (a) 8 (b) 12 (c) 6 (d) 1
Why: (a) 8 corners. (b) 12 edges, 1 middle cube each. (c) 6 faces, 1 middle cube each. (d) 1 cube inside.

11 page 37
Answer: 30 cubes
Why: 32 ÷ 6 = 5, 20 ÷ 6 = 3 and 14 ÷ 6 = 2 (whole cubes only). 5 × 3 × 2 = 30 cubes.

12 page 37
Answer: 30 cm
Why: The water has a volume of 45 × 40 × 30 = 54 000 cm³. The floor of the tank is 60 × 30 = 1800 cm². The depth is 54 000 ÷ 1800 = 30 cm.

13 page 37
Answer: 168 litres
Why: 7/12 - 3/8 = 14/24 - 9/24 = 5/24 of the tank is 35 litres. So 1/24 is 7 litres and the full tank is 24 × 7 = 168 litres.

14 page 37
Answer: (a) True (b) False (c) False
Why: (a) True. (b) The cuboid is 24 cm³. The cube is 27 cm³, so it is false. (c) False: 2 × 2 × 2 = 8, so the volume is 8 times as much.

THINK HARDER

1 page 38
Answer: 216 small cubes
Why: 64 = 4 × 4 × 4 unpainted cubes inside. The painted layer adds 1 on each side: 4 + 2 = 6 across. 6 × 6 × 6 = 216.

2 page 38
Answer: 16 cm
Why: The water is 50 × 20 × 24 = 24 000 cm³. The floors are 1000 + 500 = 1500 cm². 24 000 ÷ 1500 = 16 cm.

3 page 38
Answer: 10 cm
Why: Water 50 × 30 × 8 = 12 000 cm³. Free floor 1500 - 300 = 1200 cm². 12 000 ÷ 1200 = 10 cm (the block is 12 cm tall, so this works).

GO FURTHER

1 page 39
Answer: 160 cm³
Why: The cube is 216 small cubes. Each tunnel has 24. The 8 middle cubes are in all 3 tunnels, so the tunnels remove 72 - 16 = 56. 216 - 56 = 160 cm³.

2 page 39
Answer: 15 minutes
Why: The tank holds 60 × 50 × 40 = 120 000 cm³ = 120 litres. Each minute the tank gains 9 - 1 = 8 litres. 120 ÷ 8 = 15 minutes.

3 page 39
Answer: 140 cm³
Why: Try edges 4, 5 and 7: 4 × 5 = 20, 4 × 7 = 28 and 5 × 7 = 35. They fit. The volume is 4 × 5 × 7 = 140 cm³.

CHECK YOURSELF

1 page 40
Answer: 240 cm³
Why: 12 × 5 × 4 = 60 × 4 = 240 cm³.

2 page 40
Answer: 12 cm
Why: 18 litres = 18 000 cm³. The base is 50 × 30 = 1500 cm². The depth is 18 000 ÷ 1500 = 12 cm.

3 page 40
Answer: 36 cubes
Why: The cube has 12 edges. On each edge the 3 middle cubes have 2 painted faces (the 2 end cubes are corners). 12 × 3 = 36.

4 page 40
Answer: 1200 cm³
Why: The rise is 14 - 10 = 4 cm. The volume is 20 × 15 × 4 = 300 × 4 = 1200 cm³.

Unit 6 · Rates and deals

Your turn page 41
Answer: 66 bottles
Why: 24 ÷ 4 = 6 bottles in 1 minute. In 11 minutes: 6 × 11 = 66 bottles.

START HERE

1 page 42
Answer: 6, 30, 48
Why: In 3 minutes: 18 litres. In 1 minute: 18 ÷ 3 = 6 litres. In 5 minutes: 5 × 6 = 30 litres. In 8 minutes: 8 × 6 = 48 litres.

2 page 42
Answer: (a) 0.70 (b) 6.30
Why: (a) $4.20 ÷ 6 = $0.70. (b) 9 × $0.70 = $6.30.

3 page 42
Answer: (a) 9 (b) 225 (c) 30
Why: (a) 36 ÷ 4 = 9 rolls a minute. (b) 25 × 9 = 225. (c) 270 ÷ 9 = 30 minutes.

4 page 42
Answer: (a) 0.50 (b) 0.45 (c) Pack B
Why: (a) $2.50 ÷ 5 = $0.50. (b) $3.60 ÷ 8 = $0.45. (c) $0.45 is less than $0.50, so Pack B is the better buy.

5 page 43
Answer: (a) 12.00 (b) 11.25 (c) Shop Y
Why: 15 cans is 5 packs of 3 at X: 5 × $2.40 = $12.00. 15 cans is 3 packs of 5 at Y: 3 × $3.75 = $11.25. Shop Y is cheaper.

6 page 43
Answer: (a) 20 (b) 10
Why: (a) 14 + 6 = 20 litres a minute. (b) 200 ÷ 20 = 10 minutes.

7 page 43
Answer: 20 minutes
Why: The tank gains 25 - 10 = 15 litres each minute. 300 ÷ 15 = 20 minutes.

8 page 43
Answer: 20 minutes
Why: Together they fold 35 + 25 = 60 booklets a minute. 1200 ÷ 60 = 20 minutes.

9 page 43
Answer: $11.40
Why: First 2 hours $3.00. Then 1 hour 40 minutes is 6 quarter hours and a part, so 7 lots: 7 × $1.20 = $8.40. Total $11.40.

10 page 44
Answer: (a) 105 (b) 16
Why: (a) 9 × 15 = 135 litres are taken out. 240 - 135 = 105 litres are left. (b) 240 ÷ 15 = 16 minutes.

11 page 44
Answer: (a) 24 (b) 1440 (c) 1.44
Why: (a) 1 minute is 60 seconds, or 6 lots of 10 seconds. 6 × 4 = 24 ml. (b) 60 minutes: 60 × 24 = 1440 ml. (c) 1440 ÷ 1000 = 1.44 litres.

12 page 44
Answer: (a) True (b) False (c) True
Why: (a) True: double the cans, double the cost. (b) False: the larger pack can cost more for each item. (c) True: 5 + 7 = 12.

13 page 44
Answer: 6 visits
Why: Plan B costs $3 more a visit. The $18 fee is paid back after 18 ÷ 3 = 6 visits (both $30).

14 page 44
Answer: 22.5 litres
Why: 5 minutes is 300 seconds, which is 15 lots of 20 seconds. 15 × 1.5 = 22.5 litres.

THINK HARDER

1 page 45
Answer: $18.50 (1 pack B and 2 packs C)
Why: Ways to make 32 pens: B + 2C = $18.50; 4B = $18.80; 4A + C = $18.90. The least is $18.50.

2 page 45
Answer: 12 minutes
Why: Pretend 60 litres. P: 3 a minute. Q: 2 a minute. Together 5 a minute. 60 ÷ 5 = 12 minutes.

3 page 45
Answer: 480 boxes
Why: 1 worker packs 400 ÷ 5 ÷ 8 = 10 boxes a minute. 8 × 6 × 10 = 480 boxes.

GO FURTHER

1 page 46
Answer: 8 litres a minute
Why: First half 60 ÷ 6 = 10 minutes. Second half 60 ÷ 12 = 5 minutes. 120 ÷ 15 = 8 litres a minute (not 9: more time is spent at the slow rate).

2 page 46
Answer: 10 minutes (40 litres each)
Why: A has 80 litres more. The gap closes by 5 + 3 = 8 a minute: 10 minutes (40 litres each).

3 page 46
Answer: 20 minutes
Why: Tap A alone: 8 × 9 = 72 litres. 240 left at 20 a minute: 12 minutes. 8 + 12 = 20 minutes.

CHECK YOURSELF

1 page 47
Answer: 144 copies
Why: 54 ÷ 3 = 18 copies in 1 minute. 8 × 18 = 144 copies.

2 page 47
Answer: (a) 8 mangoes (b) 0.75
Why: 5 mangoes: $4.00 ÷ 5 = $0.80 each. 8 mangoes: $6.00 ÷ 8 = $0.75 each. $0.75 is less, so 8 mangoes is the better buy.

3 page 47
Answer: 10 minutes
Why: The tank gains 9 + 11 - 4 = 16 litres each minute. 160 ÷ 16 = 10 minutes.

4 page 47
Answer: $15.00
Why: First 30 minutes $5.00. 85 minutes is 5 lots of 20: 5 × $2.00 = $10.00. Total $15.00.

Unit 7 · Out of 100

Your turn page 48
Answer: $65.40
Why: 10% of $75 = $7.50, so 20% = $15 off. The cost after the discount is $60. 1% of $60 = $0.60, so 9% = $5.40. Mei pays $60 + $5.40 = $65.40.

START HERE

1 page 49
Answer: (a) 38 (b) 62
Why: (a) 38 of the 100 squares are shaded, so 38%. (b) 100 - 38 = 62 squares are not shaded, so 62%.

2 page 49
Answer: (a) 34 (b) 17 (c) 51
Why: 10% is $340 ÷ 10 = $34. 5% is half of that: $17. 15% = 10% + 5% = $34 + $17 = $51.

3 page 49
Answer: (a) 72 (b) 35
Why: (a) 18 ÷ 25 = 72 ÷ 100, so 72%. (b) 7 out of 20 = 35 out of 100 = 35%.

4 page 49
Answer: (a) 12 (b) 18 (c) 102
Why: (a) $120 ÷ 10 = $12. (b) 5% = $6, so 15% = $12 + $6 = $18. (c) $120 - $18 = $102.

5 page 50
Answer: (a) 5.76 (b) 69.76
Why: (a) 1% of $64 = $0.64. 9% = 9 × $0.64 = $5.76. (b) $64 + $5.76 = $69.76.

6 page 50
Answer: $72
Why: 20% of $150 = $30, so $120 is left. 40% of $120 = $48 for the book. $120 - $48 = $72. (Not 40% of $150!)

7 page 50
Answer: (a) 14 (b) 10 (c) 16
Why: (a) 10% = 4, 5% = 2: 12 + 2 = 14. (b) 25% = 10. (c) 40 - 14 - 10 = 16.

8 page 50
Answer: (a) 70 (b) 68 (c) Quiz 1
Why: (a) 21 ÷ 30 = 7 ÷ 10 = 70%. (b) 34 out of 50 = 68 out of 100 = 68%. (c) 70% is more than 68%, so Quiz 1.

9 page 50
Answer: (a) 27.20 (b) 24 (c) Shop Q
Why: (a) 2 caps $32, 15% off: $27.20. (b) $16 + $8 = $24. (c) Shop Q.

10 page 51
Answer: (a) 60 (b) 48 (c) No
Why: (a) $50 + $10 = $60. (b) 20% of $60 = $12, so $48. (c) No: the second 20% is of $60.

11 page 51
Answer: (a) 40 (b) 75
Why: (a) 2 out of 5 equal parts: 2/5 = 40/100 = 40%. (b) 3 out of 4 equal parts: 3/4 = 75%.

12 page 51
Answer: (a) True (b) True (c) False
Why: (a) True. (b) True. (c) False: $100 → $80 → $72, but 30% off gives $70.

13 page 51
Answer: $126
Why: 30% of $240 = $72 for the bag, so $168 is left. 25% of $168 = $42 for the book. $168 - $42 = $126.

14 page 51
Answer: $292.50
Why: 1% of $250 = $2.50, so 9% = $22.50. The sofa costs $250 + $22.50 = $272.50. Add the delivery: $272.50 + $20 = $292.50.

THINK HARDER

1 page 52
Answer: (a) 144 (b) 28
Why: 10% of 200 = 20, so 180. 20% of 180 = 36, so 144. The 2 discounts total $56 of $200 = 28%.

2 page 52
Answer: $80
Why: $60 is 75% of the old cost. 25% = $20, so 100% = $80.

3 page 52
Answer: $32
Why: $432 is 108% of the cost before GST, so 1% = $4. The GST is 8 × $4 = $32.

GO FURTHER

1 page 53
Answer: (a) 25 (b) 20
Why: (a) $15 out of the start $60 = 25%. (b) $15 out of the start $75 = 20%. The start amounts differ.

2 page 53
Answer: 26%
Why: Girls 30, boys 20. Badges: 9 + 4 = 13 of 50 = 26%.

3 page 53
Answer: 50 blue beads
Why: Red: 40% of 50 = 20, unchanged. Later red is 20% of all, so the total is 100 and blue is 80. Added: 80 - 30 = 50.

CHECK YOURSELF

1 page 54
Answer: (a) 18 (b) 9 (c) 27
Why: 10% = $180 ÷ 10 = $18. 5% = half of $18 = $9. 15% = $18 + $9 = $27.

2 page 54
Answer: $32.70
Why: 25% of $40 = $10, so the new cost is $30. 1% of $30 = $0.30, so 9% = $2.70. $30 + $2.70 = $32.70.

3 page 54
Answer: $30
Why: 25% of $80 = $20, so $60 is left. 50% of $60 = $30 for the gift. $60 - $30 = $30 is left.

4 page 54
Answer: Test 1
Why: 33 out of 50 = 66 out of 100 = 66%. 16 out of 25 = 64 out of 100 = 64%. 66% is higher, so Test 1.

Unit 8 · Chase the angle

Your turn page 55
Answer: 72°
Why: XY = XZ, so the angles at Y and Z are equal: angle Z = 54°. The angles of a triangle add up to 180°. Angle X = 180° - 54° - 54° = 72°.

START HERE

1 page 56
Answer: (a) 60 (b) 120 (c) 60
Why: (a) A straight line: 3 parts = 180°, so AOC = 60°. (b) BOC = 2 × 60° = 120°. (c) Vertically opposite AOC: 60°.

2 page 56
Answer: (a) 48 (b) 40
Why: (a) Angles on a straight line: 180° - 47° - 85° = 48°. (b) Angles at a point: 360° - 125° - 90° - 105° = 40°.

3 page 56
Answer: (a) 55 (b) 52
Why: Angles of a triangle add up to 180°. (a) 180° - 58° - 67° = 55°. (b) 180° - 90° - 38° = 52°.

4 page 56
Answer: (a) 72 (b) 36
Why: (a) Equal sides PQ = PR: equal angles R = Q = 72°. (b) Angles of a triangle: 180° - 72° - 72° = 36°.

5 page 57
Answer: (a) 110 (b) 70
Why: (a) Neighbouring angles of a parallelogram add up to 180°. B = 180° - 70° = 110°. (b) Opposite angles are equal. C = A = 70°.

6 page 57
Answer: (a) 52 (b) 104
Why: (a) AB = AD, so triangle ABD is isosceles: (180° - 76°) ÷ 2 = 52°. (b) Neighbouring angles add up to 180°: 104°.

7 page 57
Answer: (a) 80 (b) 122
Why: Angles between parallel sides add to 180°. (a) C = 180° - 100° = 80°. (b) D = 180° - 58° = 122°.

8 page 57
Answer: (a) 58 (b) 64 (c) 122 (d) 34
Why: (a) AB = AD: equal base angles, 58°. (b) Triangle ABD: 180° - 58° - 58° = 64°. (c) Straight line: 180° - 58° = 122°. (d) Triangle ADC: 180° - 122° - 24° = 34°.

9 page 58
Answer: (a) 60 (b) 120
Why: (a) Equilateral: 3 equal angles, 180° ÷ 3 = 60°. (b) Angles on a straight line: 180° - 60° = 120°.

10 page 58
Answer: (a) 40 (b) 55 (c) 2
Why: (a) 180° - 70° - 70° = 40°. (b) 180° - 70° = 110°, so 55° each. (c) Both work: 2.

11 page 58
Answer: (a) 120 (b) 120 (c) 150
Why: (a) 6° a minute: 20 × 6° = 120°. (b) Hours are 30° apart: 4 × 30° = 120°. (c) 7 × 30° = 210°, so the smaller angle is 150°.

12 page 58
Answer: (a) angles of a triangle (b) angles on a straight line (c) neighbouring angles
Why: (a) 55° + 75° + 50° = 180°: a triangle. (b) A straight line: 180°. (c) A and B are neighbours.

THINK HARDER

1 page 59
Answer: (a) 110 (b) 35 (c) 65
Why: (a) 180° - 70° = 110°. (b) AD = DC: (180° - 110°) ÷ 2 = 35°. (c) C = 100°, so ACB = 100° - 35° = 65°.

2 page 59
Answer: (a) 76 (b) 44
Why: (a) 3 parts + 24° = 180°: 1 part = 52°, top 76°. (b) 3 parts + 48° = 180°: 1 part = 44°, top 44°.

3 page 59
Answer: 20°
Why: AD = AB = AE. Join D to E: angle DAE = 80° + 60° = 140°, and AED = (180° - 140°) ÷ 2 = 20° (isosceles).

GO FURTHER

1 page 60
Answer: (a) 540 (b) 108
Why: (a) 3 triangles: 3 × 180° = 540°. (b) 540° ÷ 5 = 108°.

2 page 60
Answer: 10°
Why: The hour hand is at 120° + 20 × 0.5° = 130°. The minute hand is at 20 × 6° = 120°. The difference is 10°.

3 page 60
Answer: 36°
Why: Let angle A be 1 unit. DCA = 1, BDC = 2, DBC = 2, ACB = ABC = 2, so DCB = 1. Triangle DBC: 5 units = 180°, so 1 unit = 36°.

CHECK YOURSELF

1 page 61
Answer: 47°
Why: AOB is a straight line, so COB = 180° - 133° = 47° (angles on a straight line).

2 page 61
Answer: 40°
Why: The angles of a triangle add up to 180°. Q and R are equal. 180° - 100° = 80°, and 80° ÷ 2 = 40°.

3 page 61
Answer: 115°
Why: Opposite angles are equal, so D = B = 115°.

4 page 61
Answer: 38°
Why: Straight line: ACD = 180° - 76° = 104°. CA = CD, so (180° - 104°) ÷ 2 = 38°.

Unit 9 · Rules for patterns

Your turn page 62
Answer: 61 sticks
Why: The first hexagon needs 6 sticks. Each new hexagon shares a side, so it adds 5 sticks. Rule: sticks = 5 × hexagons + 1. For 12 hexagons: 5 × 12 + 1 = 61 sticks.

START HERE

1 page 63
Answer: (a) 21 (b) 26 (c) 5
Why: Each new hexagon shares a side, so it adds 5 sticks. (a) 21. (b) 26. (c) 5.

2 page 63
Answer: (a) 5, 1 (b) 176
Why: (a) 6 = 5 + 1 and each hexagon adds 5. (b) 5 × 35 + 1 = 176.

3 page 63
Answer: (a) 19 (b) 19, 4
Why: (a) (96 - 1) ÷ 5 = 19. (b) (100 - 1) ÷ 5 = 19 remainder 4.

4 page 63
Answer: (a) 27, 32 (b) 52 (c) No
Why: Each is 5 more. (a) 27, 32. (b) 7 + 9 × 5 = 52. (c) 93 is not a multiple of 5: no.

5 page 64
Answer: (a) 14 (b) 47 (c) 3, 2 (d) 32
Why: Figure 1 has 5 tiles, figure 2 has 8: each figure adds 3. (a) 3 × 4 + 2 = 14. (b) 3 × 15 + 2 = 47. (c) 3 and 2. (d) 96 ÷ 3 = 32.

6 page 64
Answer: (a) 7 (b) 11
Why: All small: 18 × $3 = $54, which is $28 short. Each large badge adds $4. 28 ÷ 4 = 7 large, so 11 small.

7 page 64
Answer: (a) 10 (b) 7
Why: All stools: 51 legs, 10 short. Each chair adds 1 leg: 10 chairs, 7 stools.

8 page 64
Answer: (a) 3 (b) 9
Why: All right: 60. Each wrong costs 5 + 2 = 7. (60 - 39) ÷ 7 = 3 wrong, so 9 right.

9 page 65
Answer: (a) 22 (b) 4, 2 (c) 12
Why: Each table adds 4 chairs. (a) 4 × 5 + 2 = 22. (b) 4 and 2. (c) 48 ÷ 4 = 12.

10 page 65
Answer: (a) 3, 4 (b) 40 (c) 32
Why: Each figure adds 3 and figure 1 has 7. (a) 3 and 4. (b) 40. (c) 96 ÷ 3 = 32.

11 page 65
Answer: (a) True (b) False (c) True
Why: (a) 96 ÷ 4 = 24: true. (b) 5 × 20 + 1 = 101: false. (c) They end in 6 or 1: true.

THINK HARDER

1 page 66
Answer: 6 coins
Why: All 10¢: $2.40, which is $5.40 short. Each $1 coin adds 90¢. 540 ÷ 90 = 6 coins.

2 page 66
Answer: (a) 110 (b) 12
Why: Figure 10 adds columns of 2, 4, ..., 20. Pairs make 22: 5 pairs = 110. (b) 12 × 13 = 156.

3 page 66
Answer: 625
Why: 1, 1 + 3 = 4, 1 + 3 + 5 = 9: squares. There are 25 odd numbers up to 49, so 25 × 25 = 625.

GO FURTHER

1 page 67
Answer: 1716
Why: 33 numbers, in pairs that each add to 104: 16 pairs and the middle 52. 16 × 104 + 52 = 1716.

2 page 67
Answer: (a) 10 (b) 66
Why: (a) 5 people each shake 4 hands: 20, but every handshake is counted twice, so 10. (b) 12 × 11 ÷ 2 = 66.

3 page 67
Answer: 13 right
Why: He answers 20. All right: 120. Each wrong answer costs 6 + 2 = 8. (120 - 64) ÷ 8 = 7 wrong, so 13 right.

CHECK YOURSELF

1 page 68
Answer: 76
Why: Each number is 5 more. There are 14 jumps of 5 from the 1st to the 15th: 6 + 14 × 5 = 76.

2 page 68
Answer: Figure 12
Why: 51 - 3 = 48, and 48 ÷ 4 = 12.

3 page 68
Answer: 10 chairs
Why: Suppose all 20 are stools: 60 legs. That is 70 - 60 = 10 legs short. Each chair adds 1 leg, so there are 10 chairs (and 10 stools).

4 page 68
Answer: 25 hexagons
Why: 130 - 1 = 129, and 129 ÷ 5 = 25 with 4 left over. So 25 hexagons use 126 sticks, and 4 sticks are left.

Unit 10 · Pick a start

Your turn page 69
Answer: Kumar
Why: Ben cannot have juice or milk, so Ben has tea. Kumar does not choose milk and tea is taken, so Kumar has juice. Aisha has the milk. All clues are true.

START HERE

1 page 70
Answer: 10
Why: Work backwards, undoing the last step first. Undo halve: 19 × 2 = 38. Undo add 8: 38 - 8 = 30. Undo multiply by 3: 30 ÷ 3 = 10.

2 page 70
Answer: $28
Why: Before the $6: $21 = 3/4 of the money. 1/4 = $7, so $28.

3 page 70
Answer: Siti
Why: Mei finished after Siti, and Ali did not win. So Siti won.

4 page 70
Answer: (a) 47 (b) 28
Why: Take away the extra $19: 75 - 19 = 56. Share equally: $28 each. Add the $19 back to Kumar: $47.

5 page 70
Answer: (a) 9 (b) 4 (c) 1 (d) 14
Why: (a) 9 small squares. (b) A 2 by 2 square has 2 places across and 2 up: 2 × 2 = 4. (c) 1. (d) 9 + 4 + 1 = 14.

6 page 71
Answer: Lin
Why: If Lin ate it: 1 truth (Mo). If Kai ate it: 2 truths. If Mo ate it: 2 truths. Only Lin fits.

7 page 71
Answer: 20 years
Why: The age difference stays 28. When Dev is half as old, he is 28: 20 years from now.

8 page 71
Answer: 36 stickers
Why: Work backwards. The 12 left are half of the rest, so the rest was 24. The 24 is 2/3 of his stickers, so 1/3 is 12 and he had 36.

9 page 71
Answer: (a) 8 (b) 12
Why: All hens: 40 legs, 16 short. Each cow adds 2: 8 cows, 12 hens.

10 page 72
Answer: (a) 8 and 9
Why: Pairs with product 72 have sums 73, 38, 27, 22, 18, 17. Only 8 and 9 give 17.

11 page 72
Answer: (a) 6 (b) 132 and 312 (c) 231
Why: 123, 132, 213, 231, 312, 321: 6. (b) Even: 132 and 312. (c) The differences from 250 are 127, 118, 37, 19, 62, 71: 231.

12 page 72
Answer: 18 times
Why: Ones place: 7, 17, ..., 77 is 8. Tens place: 70 to 79 is 10. Total 18.

THINK HARDER

1 page 73
Answer: (a) swimming (b) badminton (c) tennis (d) football
Why: Clue 3: Wei plays football. Ben (not tennis, not badminton) plays swimming. Kumar (not swimming, not tennis) plays badminton. Aisha plays tennis.

2 page 73
Answer: (a) 14 (b) 45
Why: (a) 10 + 4 = 14. (b) Height 1: 10 + 8 + 6 + 4 + 2 = 30. Height 2: 5 + 4 + 3 + 2 + 1 = 15. Total 45.

3 page 73
Answer: 24 customers
Why: After the last step 14 = 2/3 of the number before, so that was 21. Before 3 arrived: 18. 18 = 3/4 of the start, so 1/4 = 6 and the start was 24.

GO FURTHER

1 page 74
Answer: 24 numbers
Why: Sets with sum 8: 0 1 7, 0 2 6, 0 3 5 (4 numbers each) and 1 2 5, 1 3 4 (6 each). 12 + 12 = 24.

2 page 74
Answer: (a) 9 (b) 9 (c) 12
Why: A 5¢ and 20¢ pair is 25¢. All 10¢ is $3.00, 45¢ short. A pair in place of two 10¢ coins adds 5¢: 9 pairs. 30 - 18 = 12 coins of 10¢.

3 page 74
Answer: 4 socks
Why: Worst case: 1 red, 1 blue, 1 black, no pair. The 4th sock matches one.

CHECK YOURSELF

1 page 75
Answer: 8
Why: Work backwards. Undo multiply by 4: 60 ÷ 4 = 15. Undo add 7: 15 - 7 = 8.

2 page 75
Answer: $35
Why: Before the $9 she had $12 + $9 = $21. The $21 is 3/5 of her money, so 1/5 is $7 and the whole is 5 × $7 = $35.

3 page 75
Answer: 4 goats
Why: Suppose all 9 are hens: 18 legs. That is 26 - 18 = 8 legs short. Each goat adds 2 legs, so 8 ÷ 2 = 4 goats.

4 page 75
Answer: 8 squares
Why: There are 6 small squares. A 2 by 2 square can go in 2 places. 6 + 2 = 8.